

MTH104 MID TERM BY IQRA AZHAR

MTH104

Mid term

2023

Solved by



Iqra Azhar



YTUBE



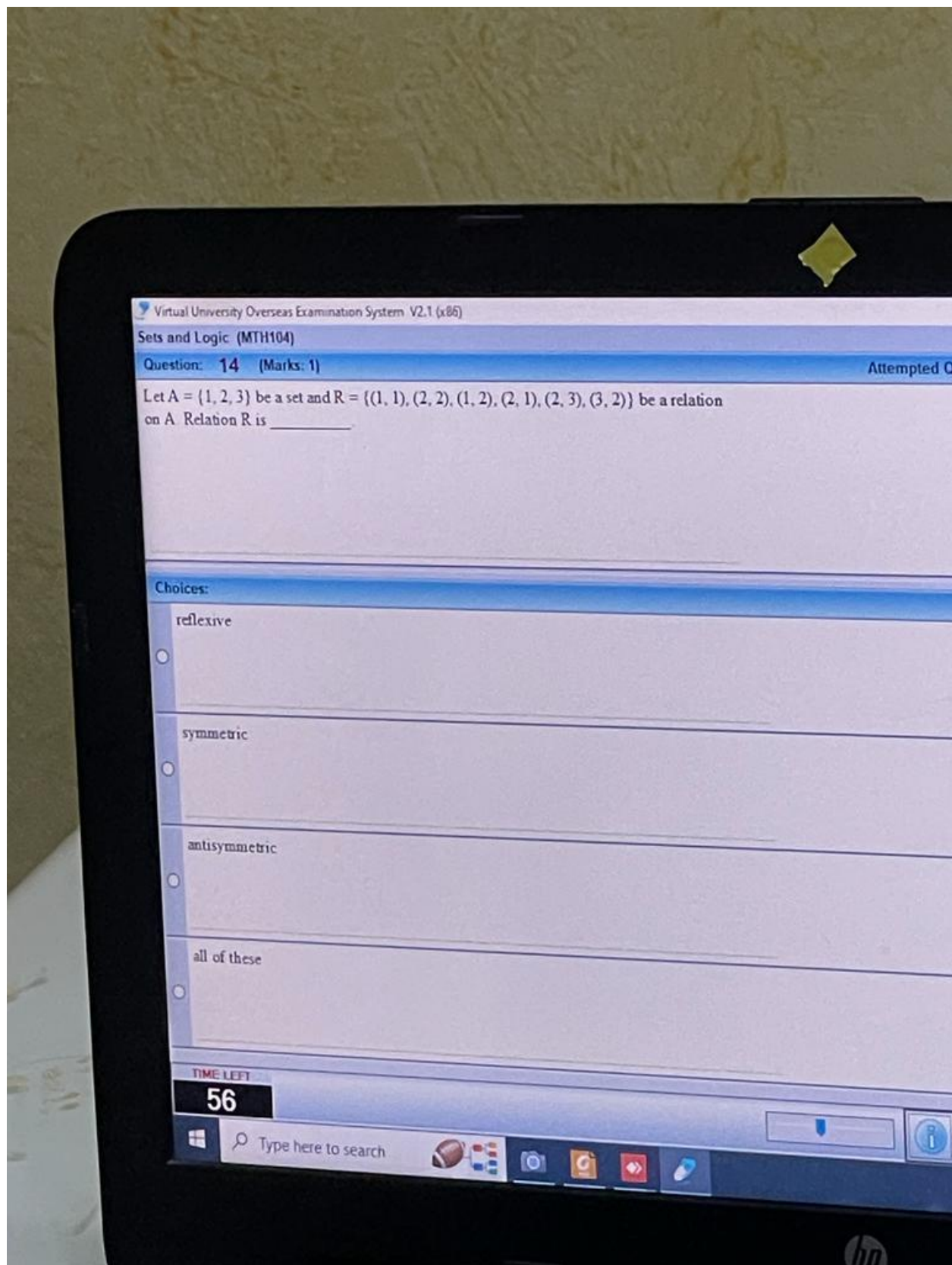
MATH ZONE FOR VU



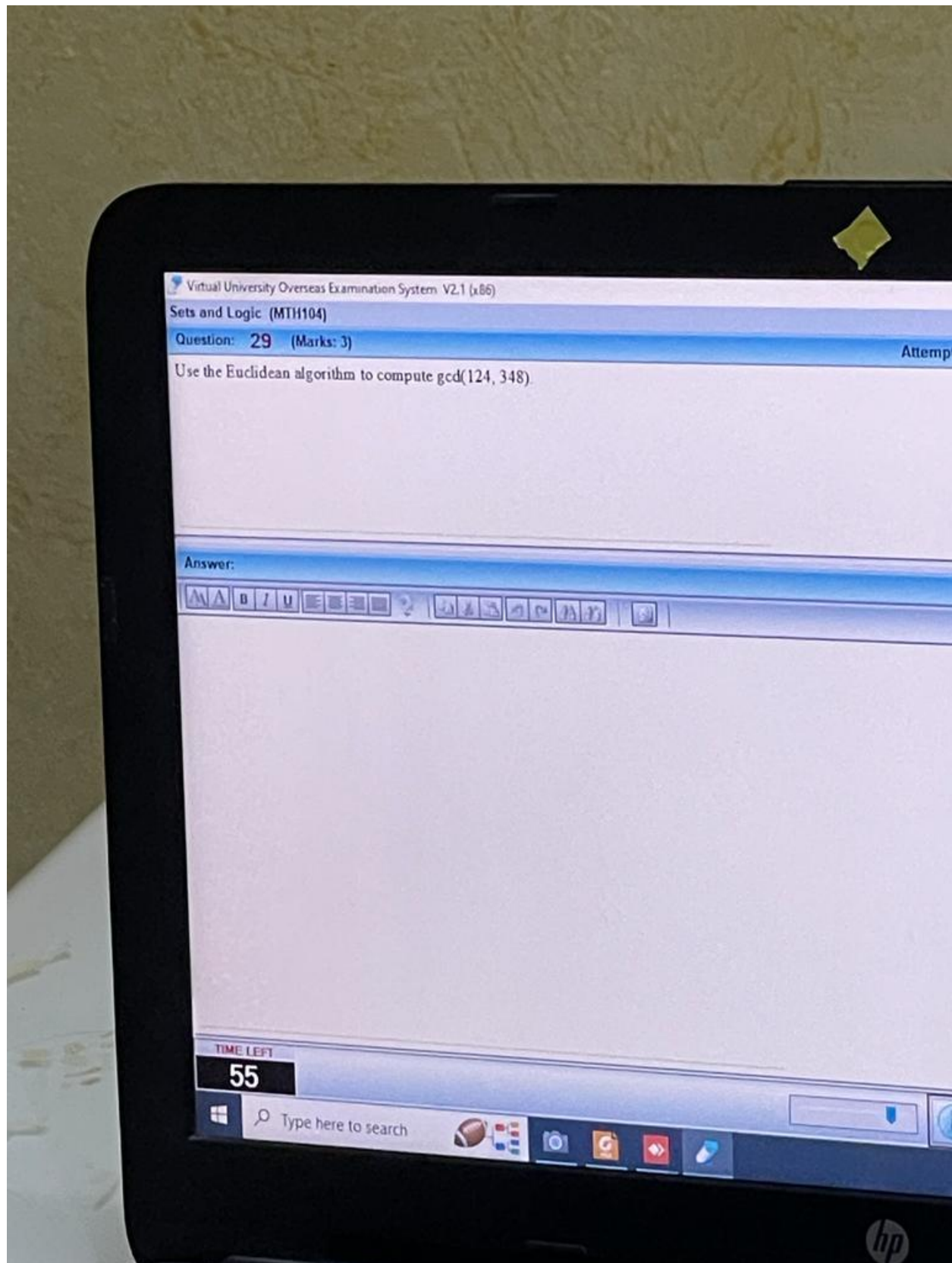
Assignments || Quiz || Gdb || Projects
Seminar || online book shop || Lms
handling services ||available

CONTACT US 03235429064

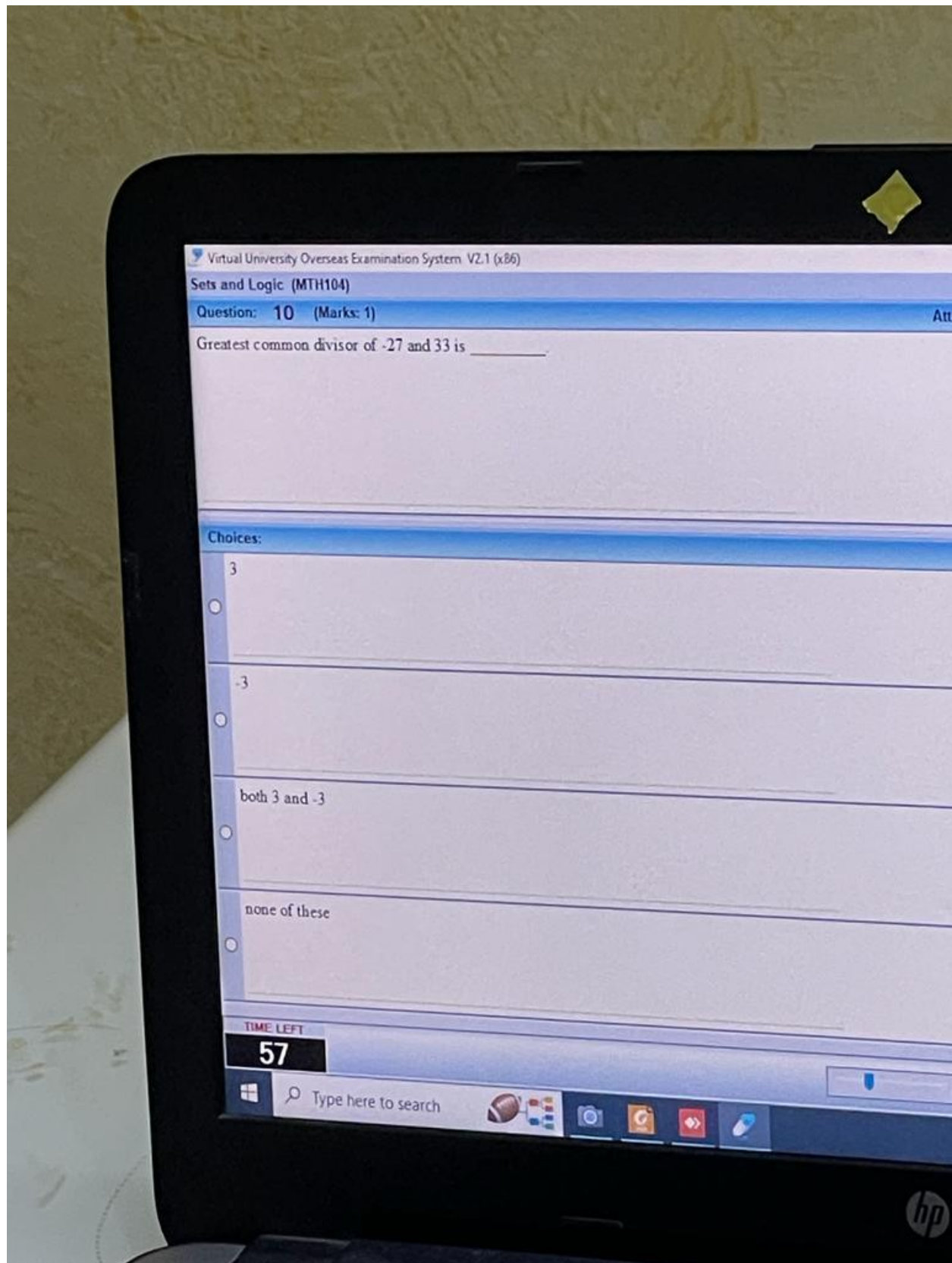
MTH104 MID TERM BY IQRA AZHAR



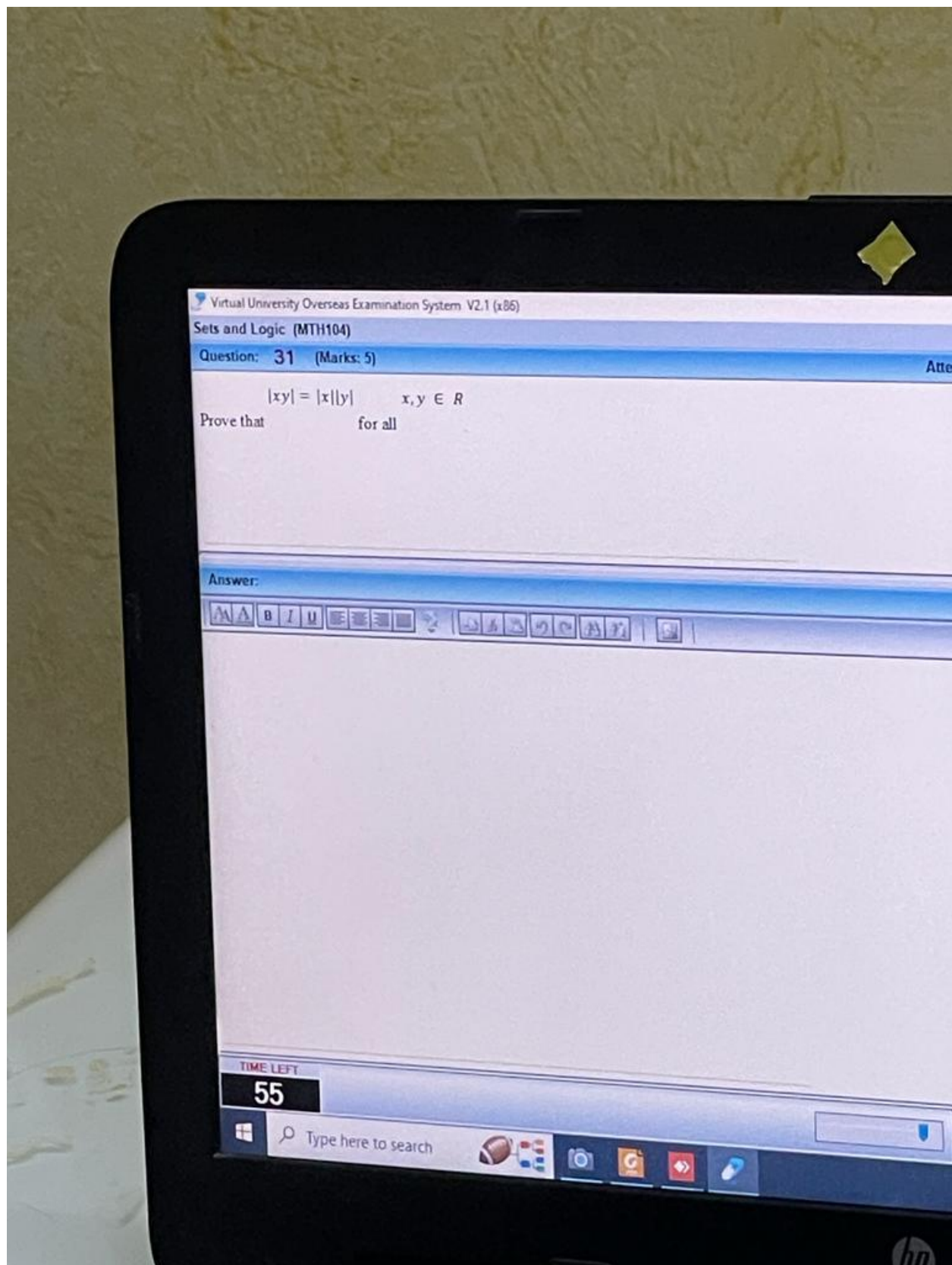
MTH104 MID TERM BY IQRA AZHAR



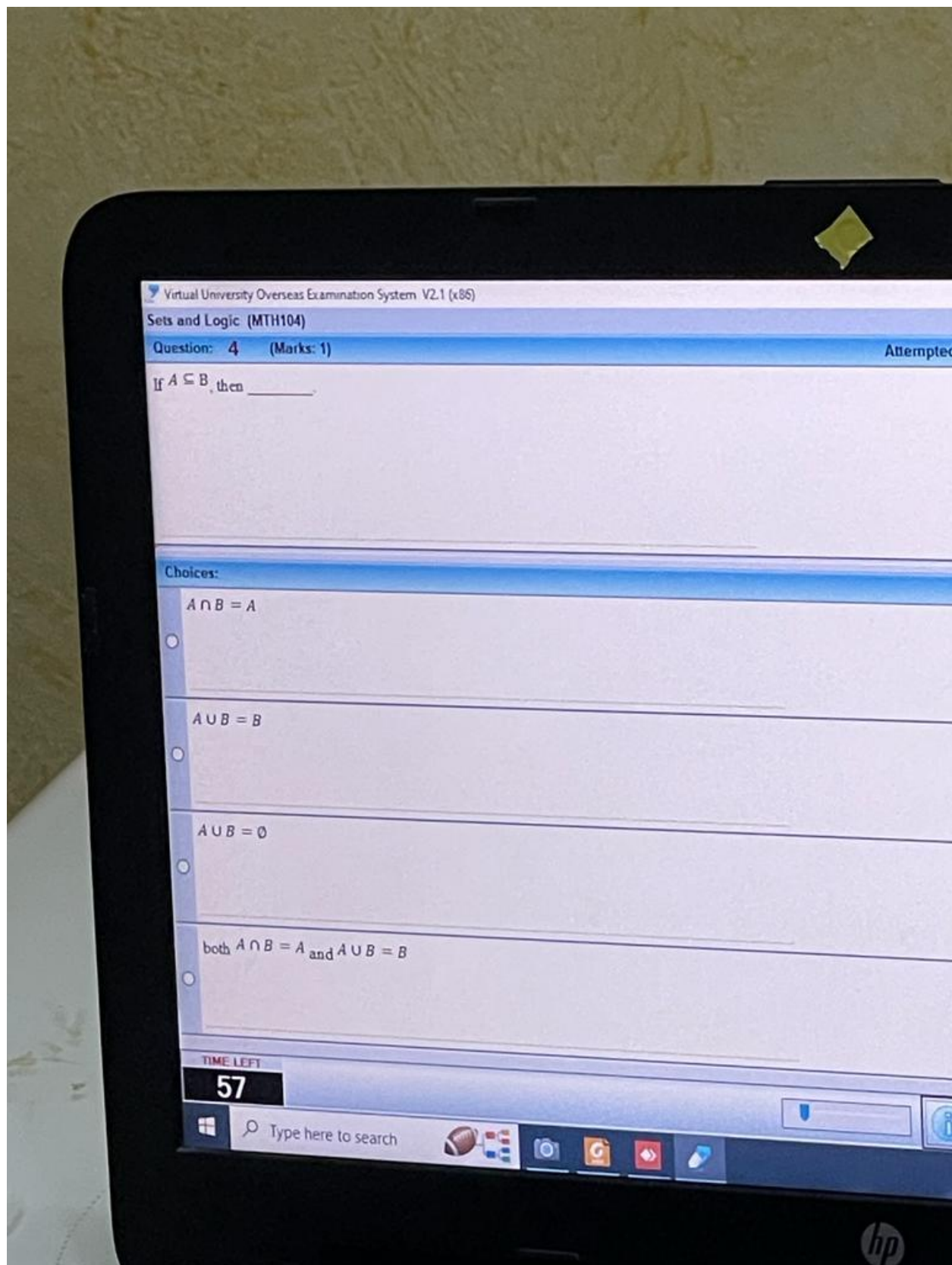
MTH104 MID TERM BY IQRA AZHAR



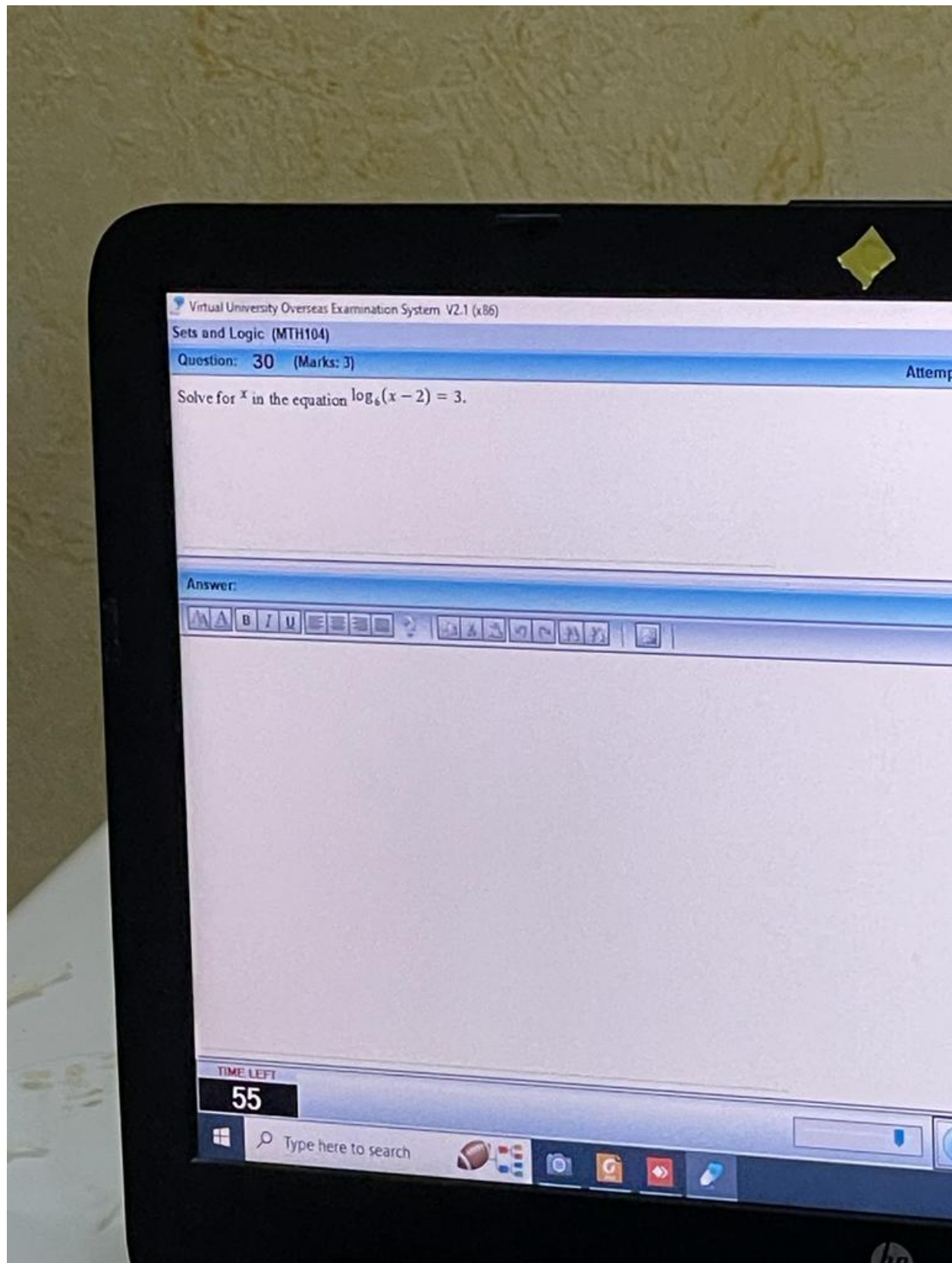
MTH104 MID TERM BY IQRA AZHAR



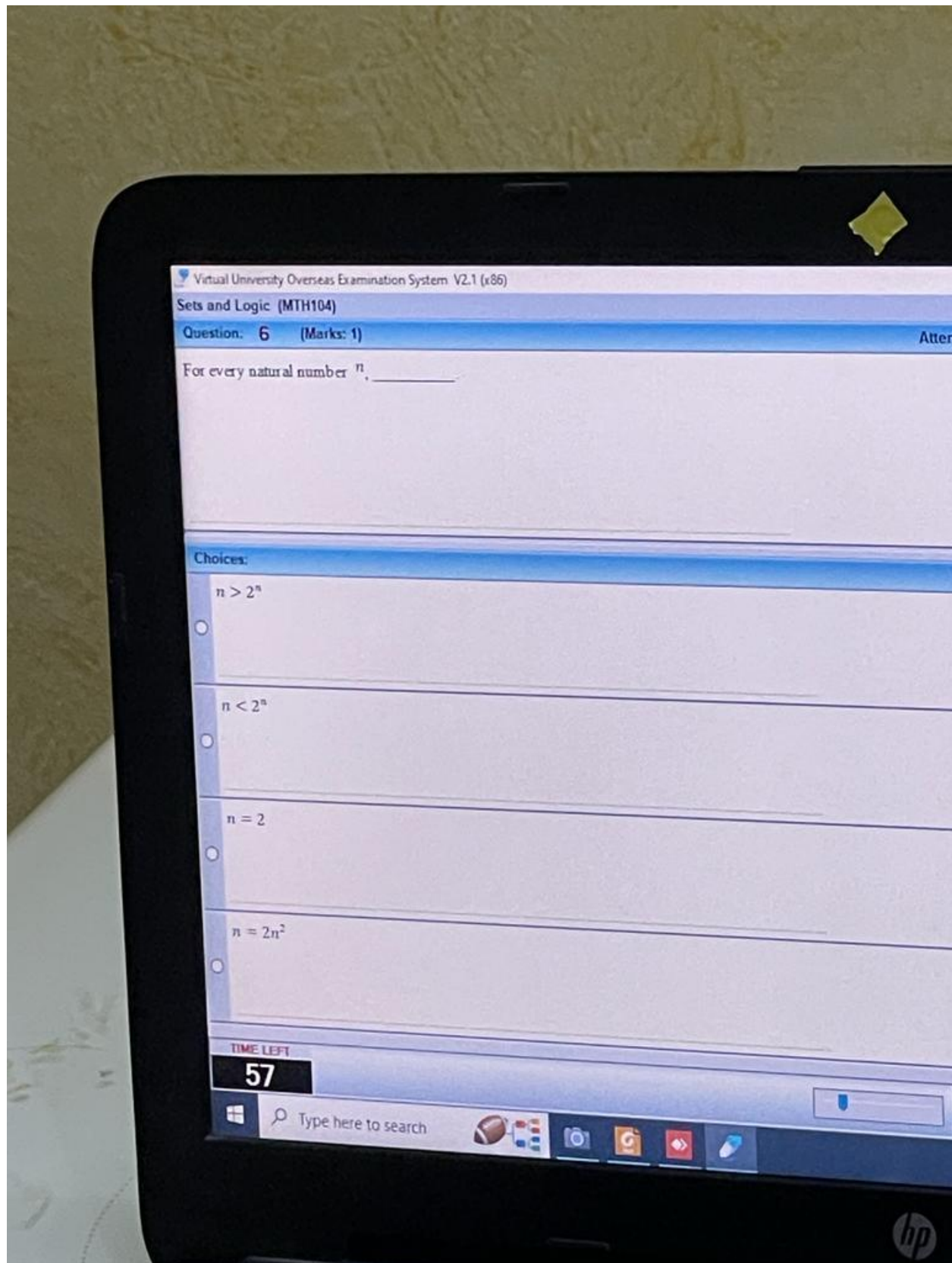
MTH104 MID TERM BY IQRA AZHAR



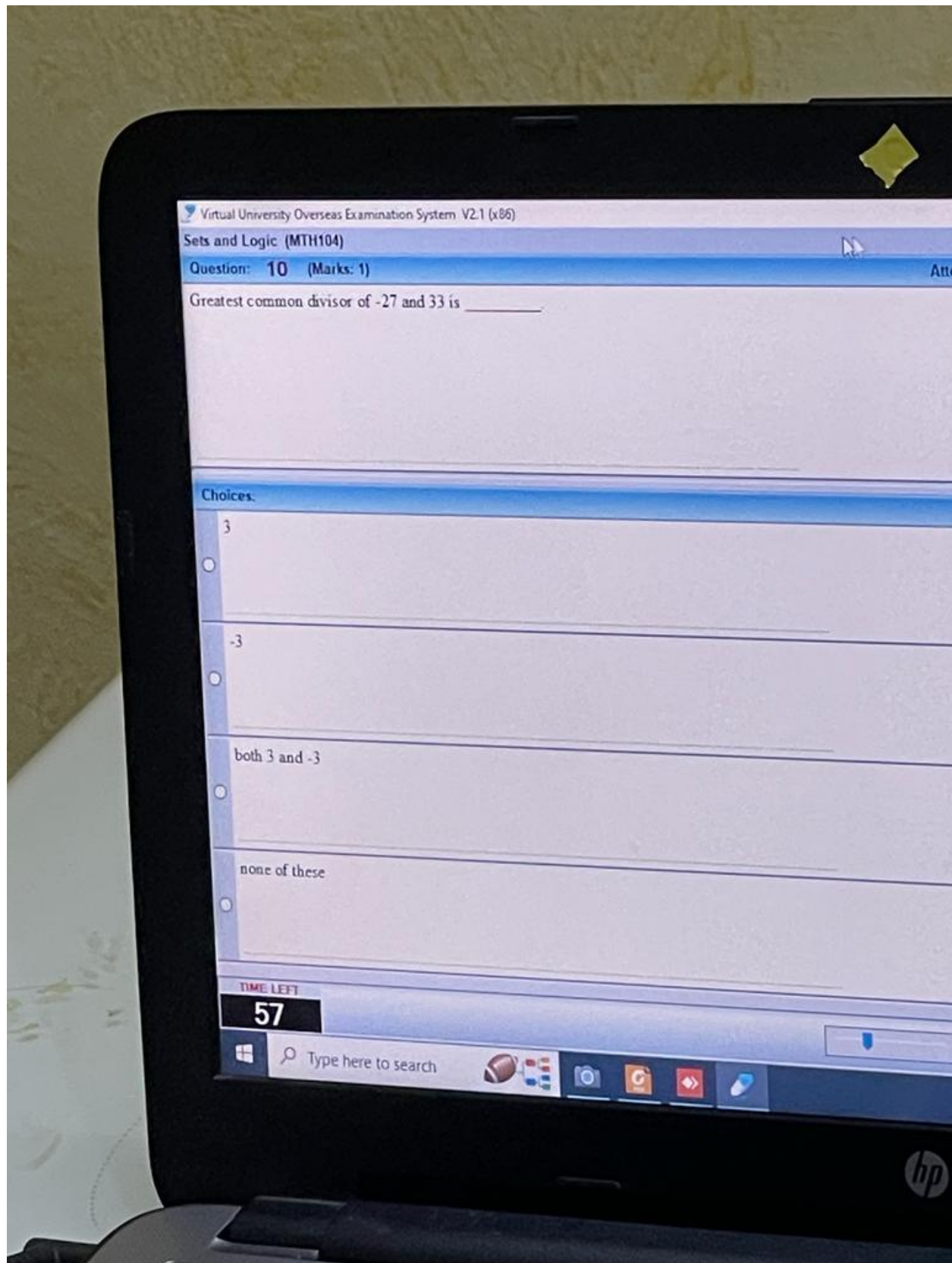
MTH104 MID TERM BY IQRA AZHAR



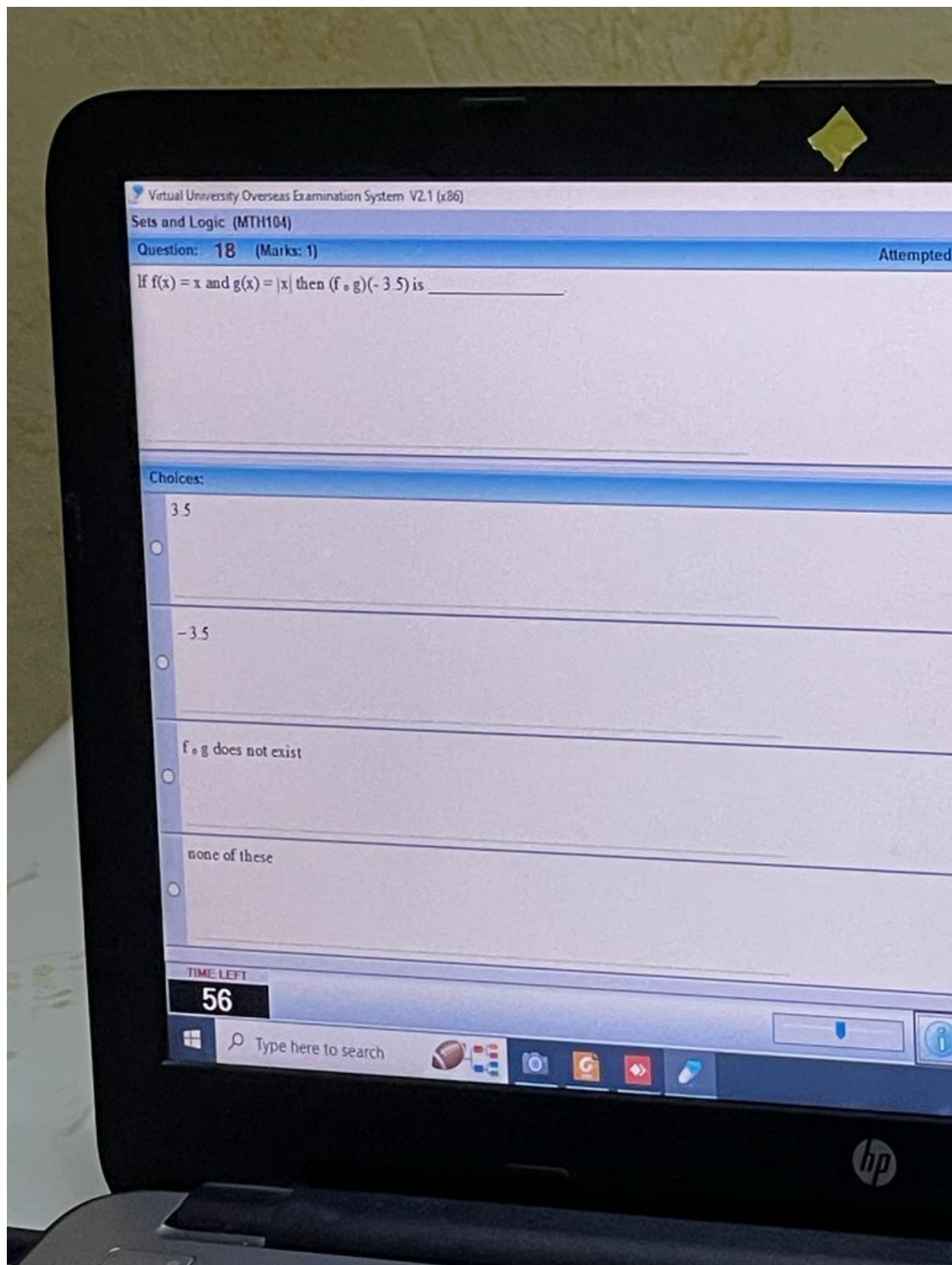
MTH104 MID TERM BY IQRA AZHAR



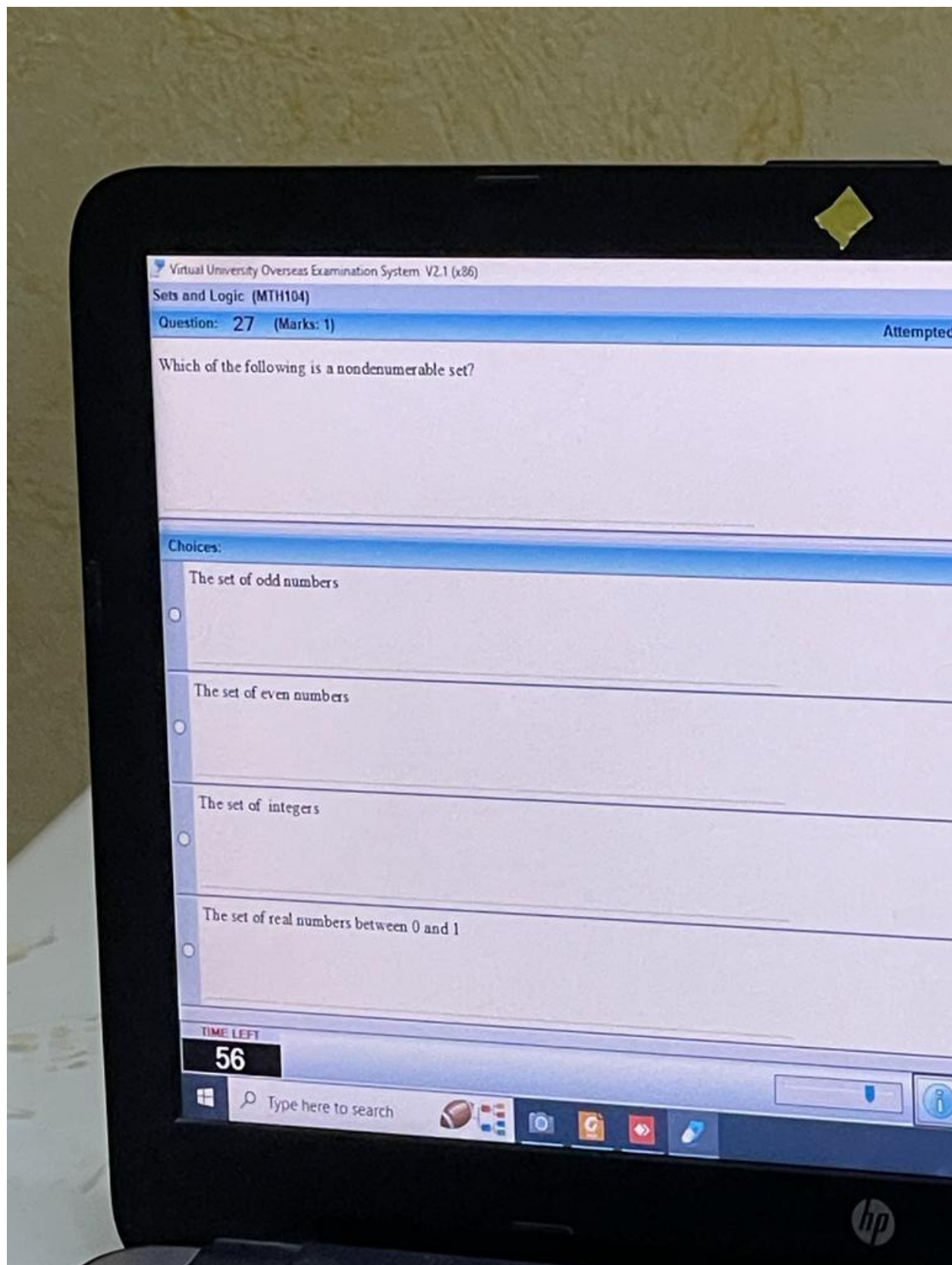
MTH104 MID TERM BY IQRA AZHAR



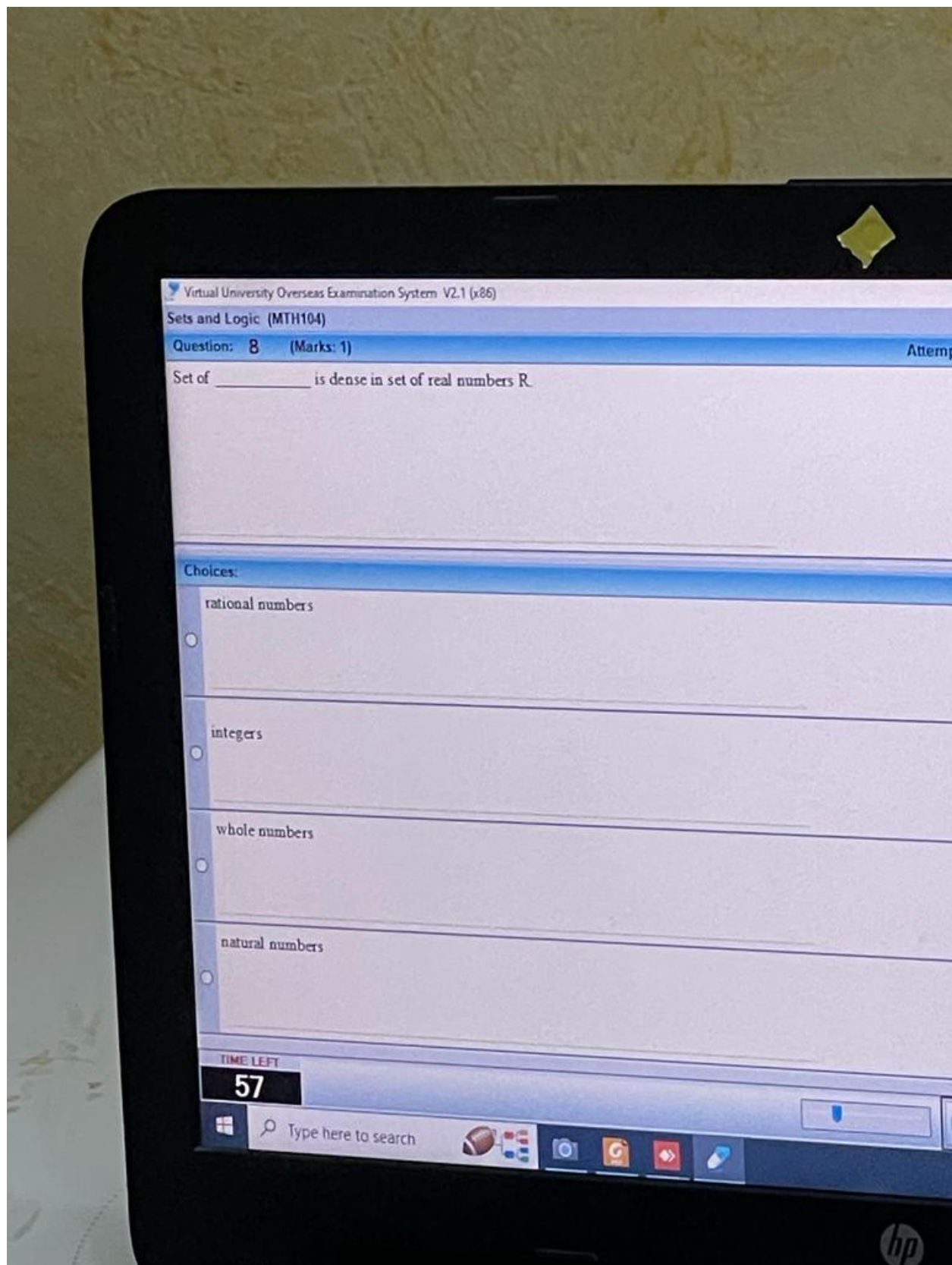
MTH104 MID TERM BY IQRA AZHAR



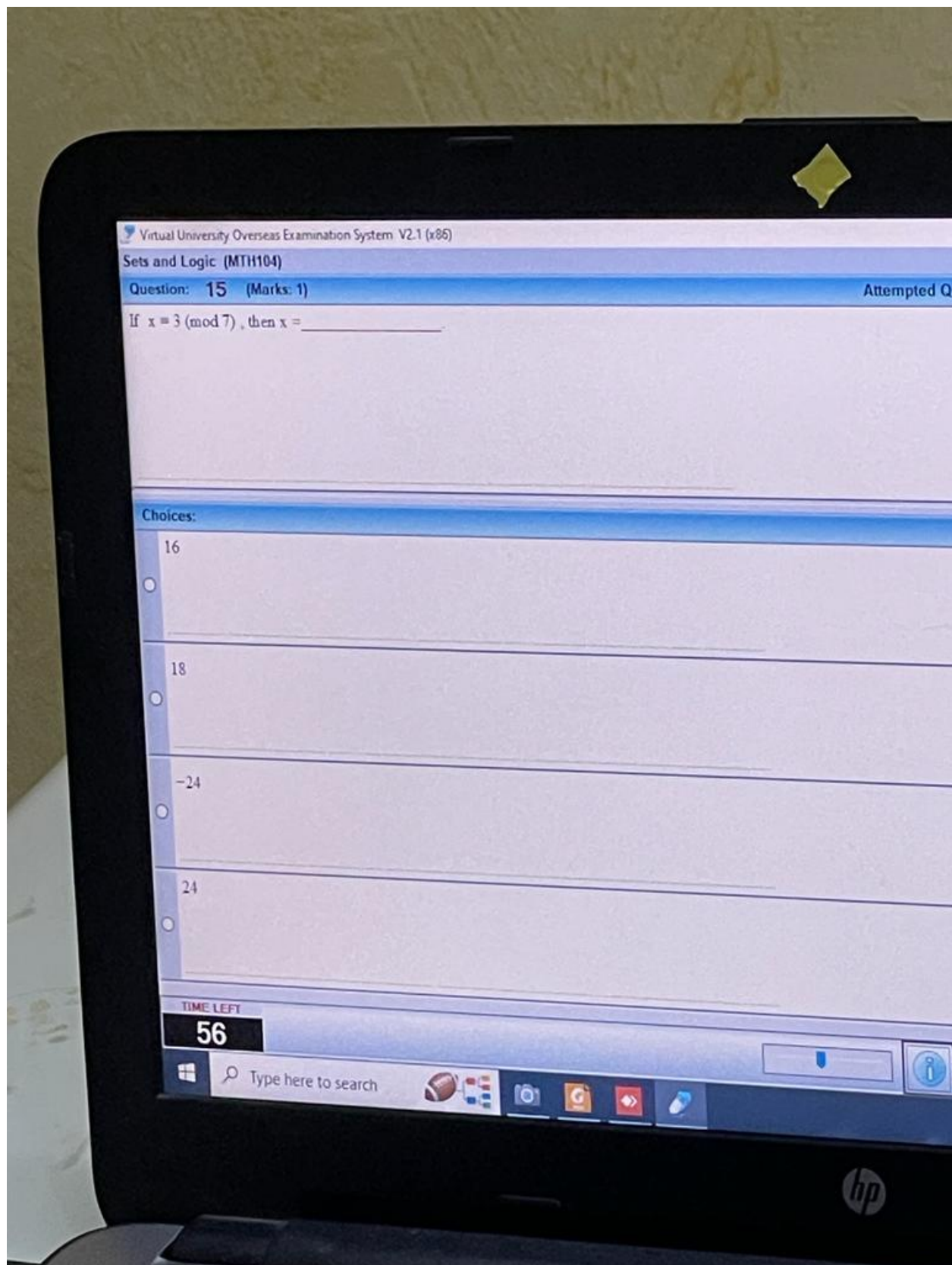
MTH104 MID TERM BY IQRA AZHAR



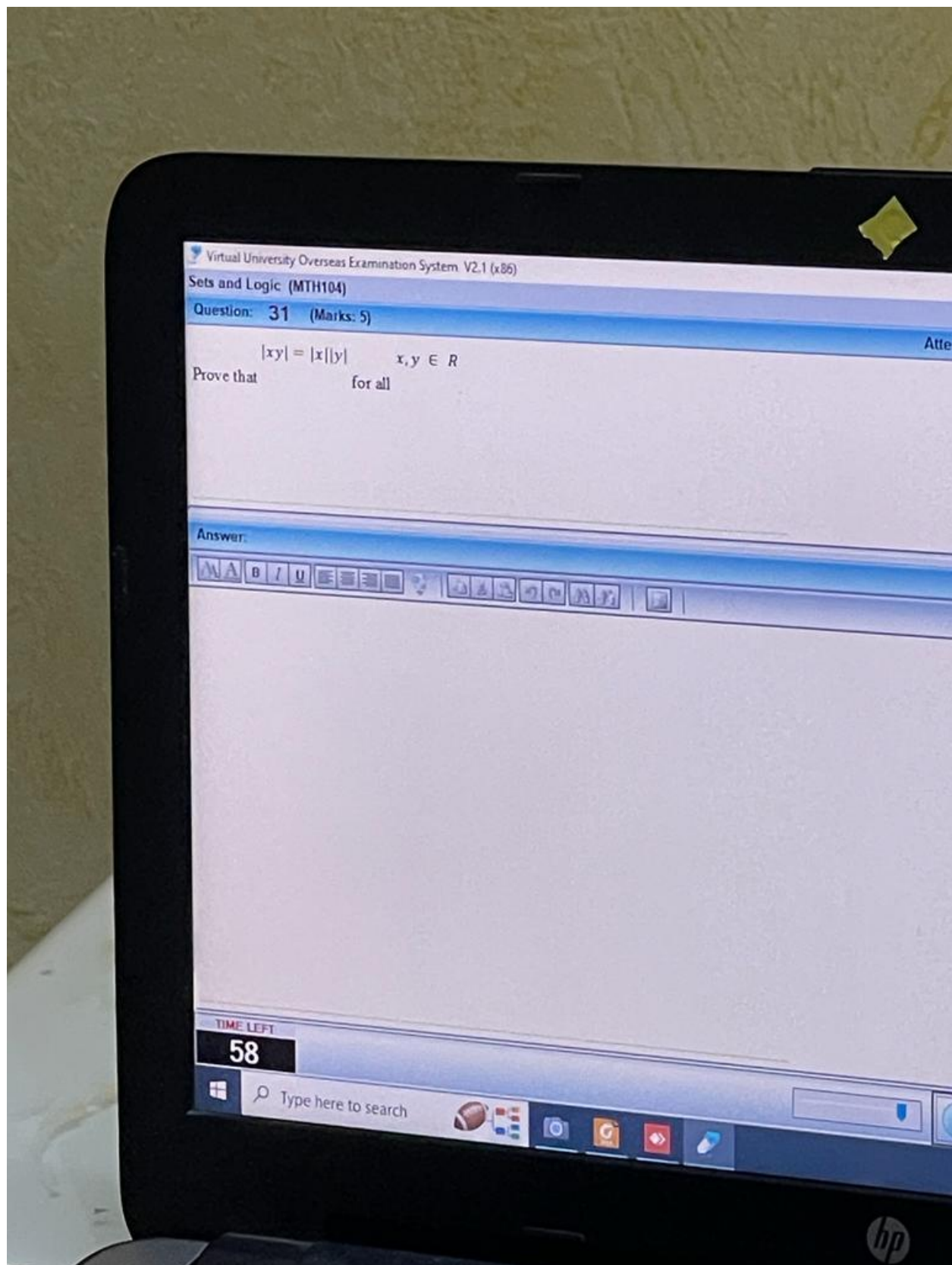
MTH104 MID TERM BY IQRA AZHAR



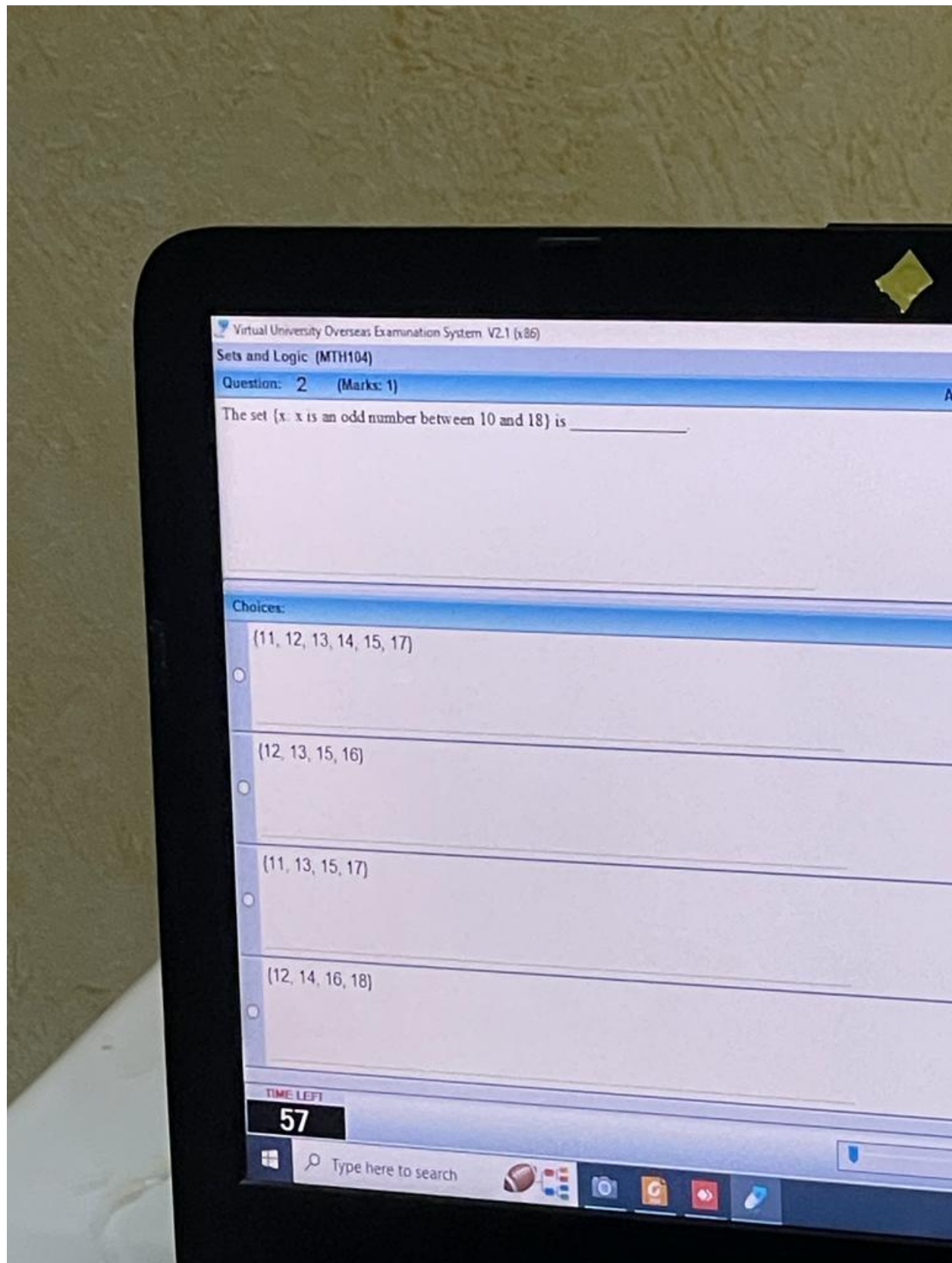
MTH104 MID TERM BY IQRA AZHAR



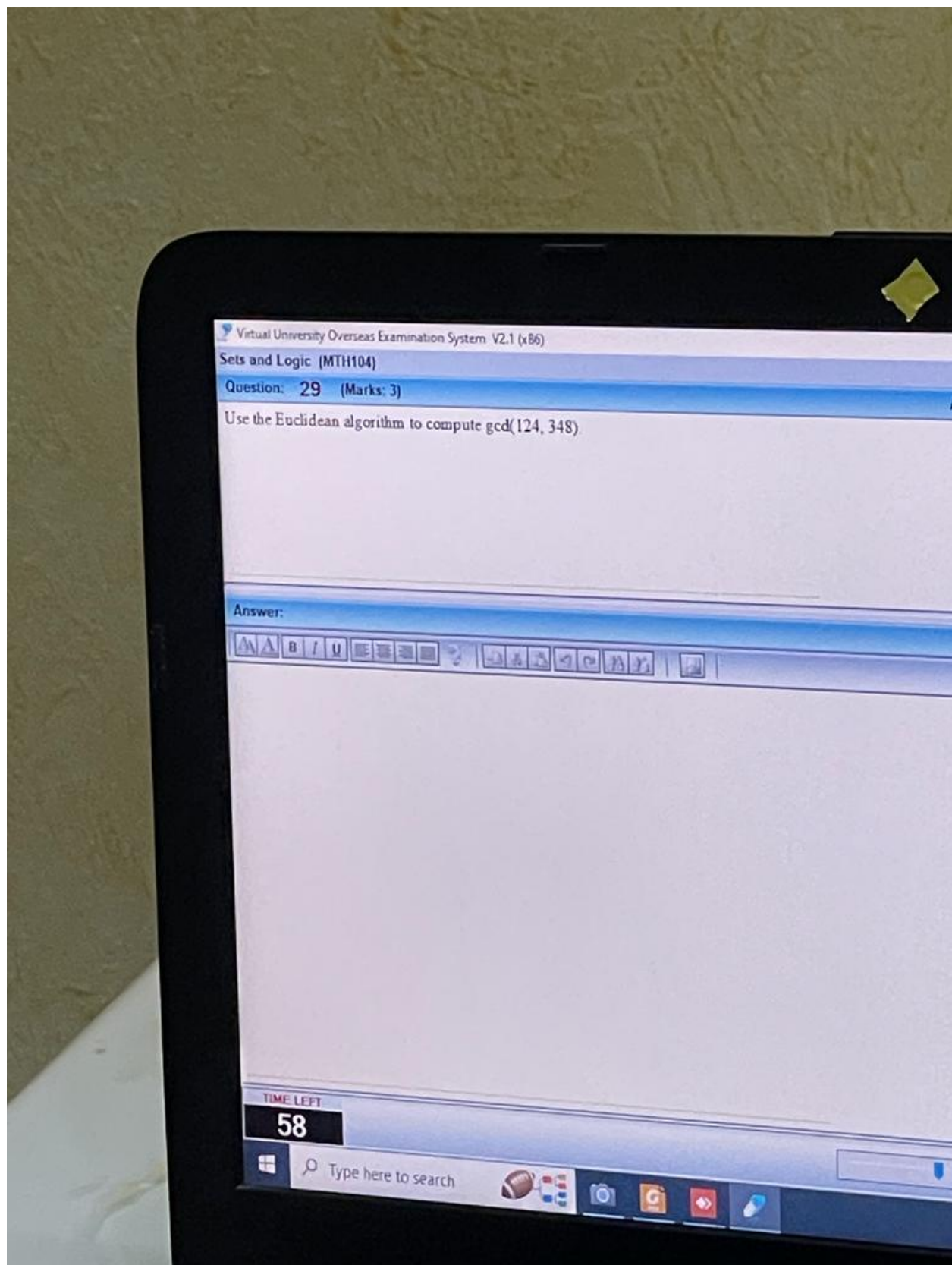
MTH104 MID TERM BY IQRA AZHAR



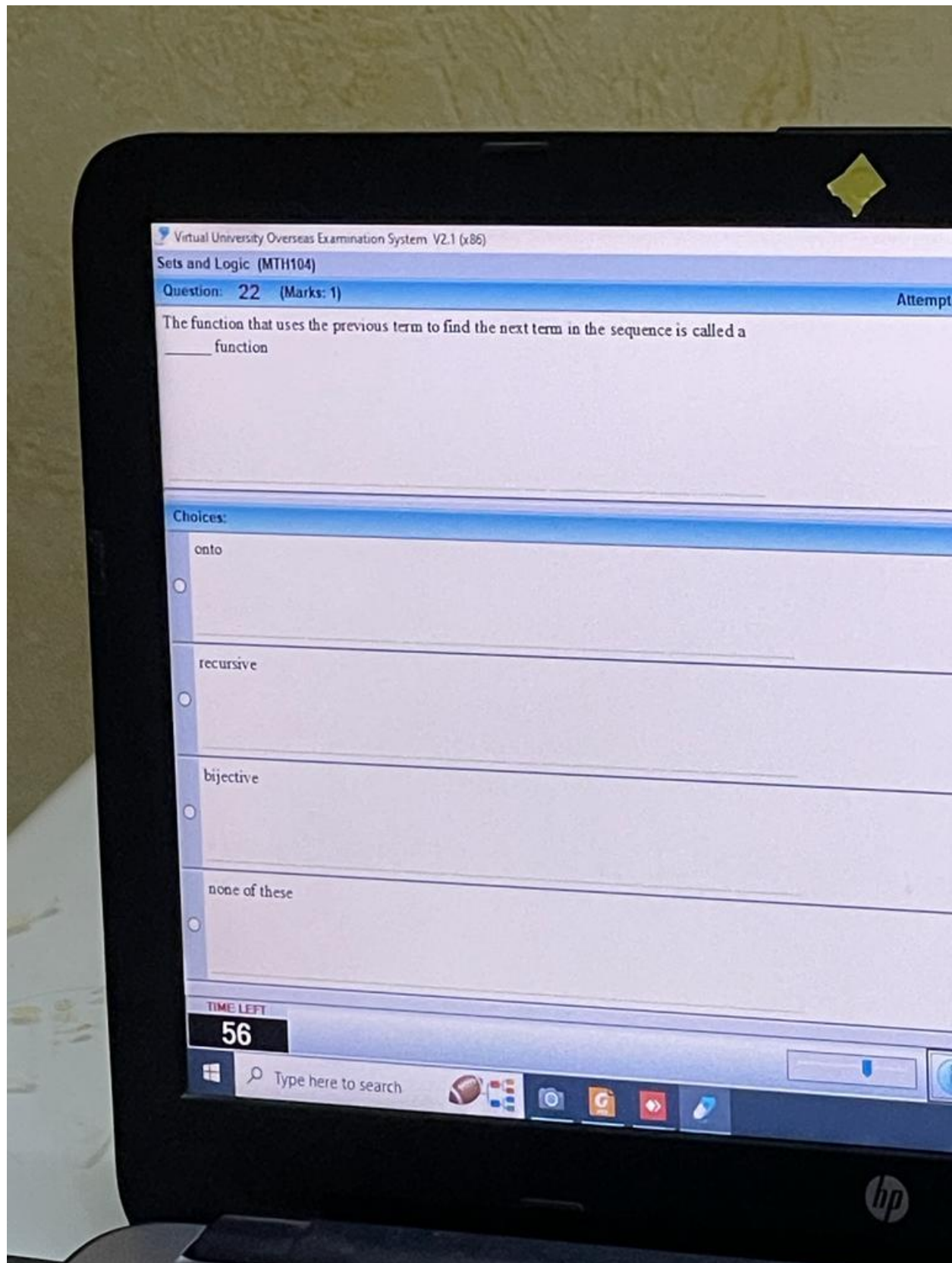
MTH104 MID TERM BY IQRA AZHAR



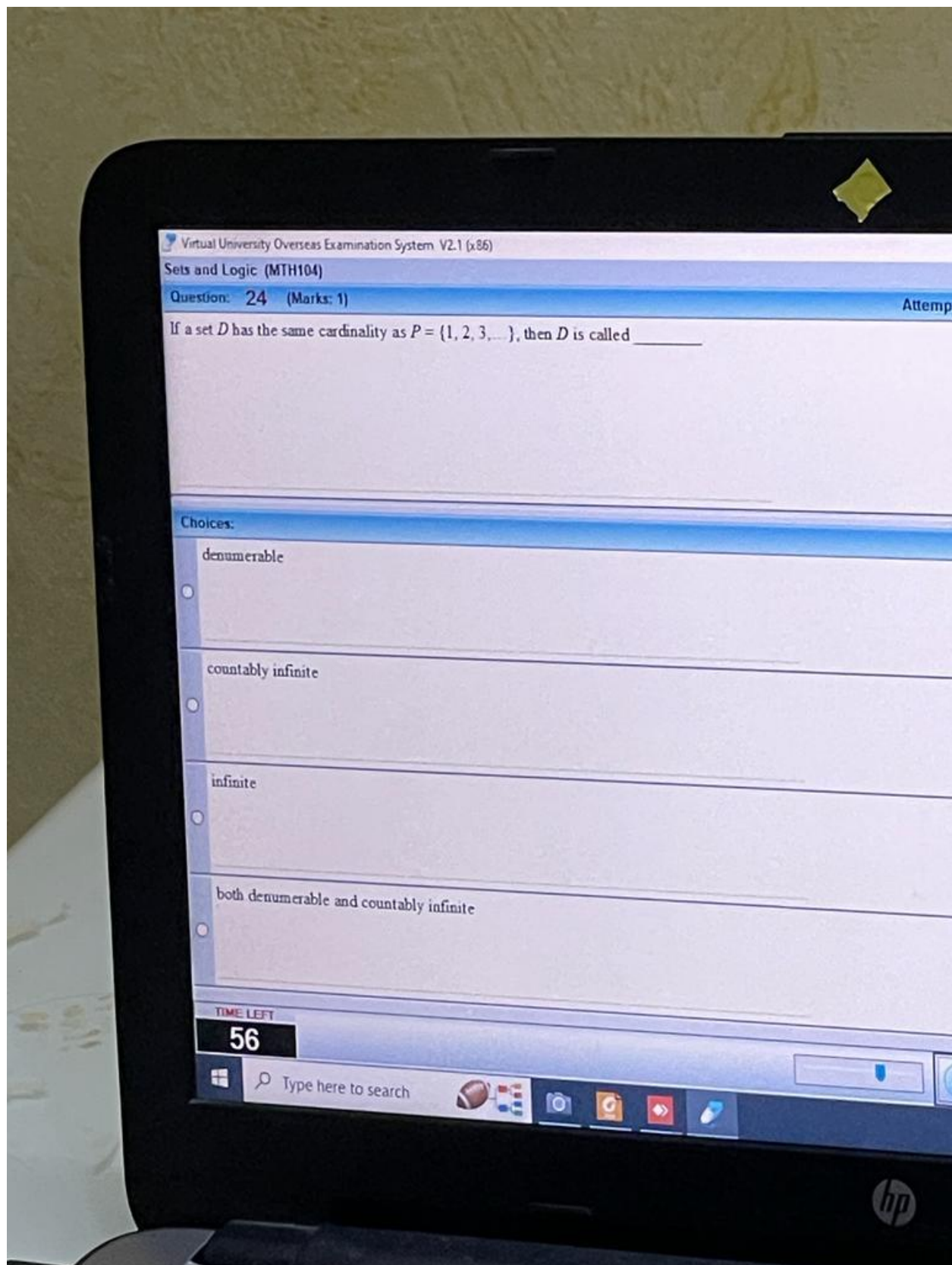
MTH104 MID TERM BY IQRA AZHAR



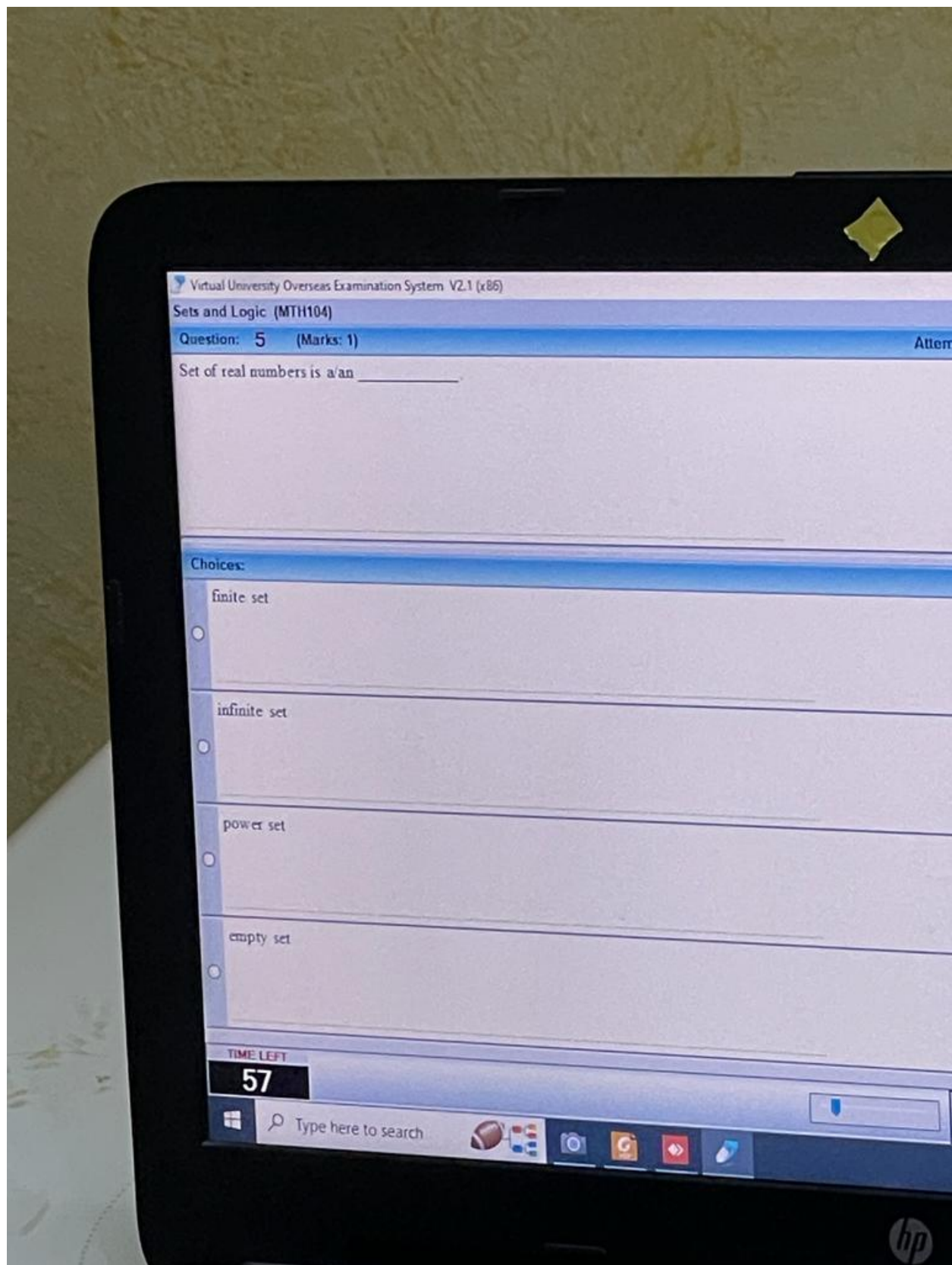
MTH104 MID TERM BY IQRA AZHAR



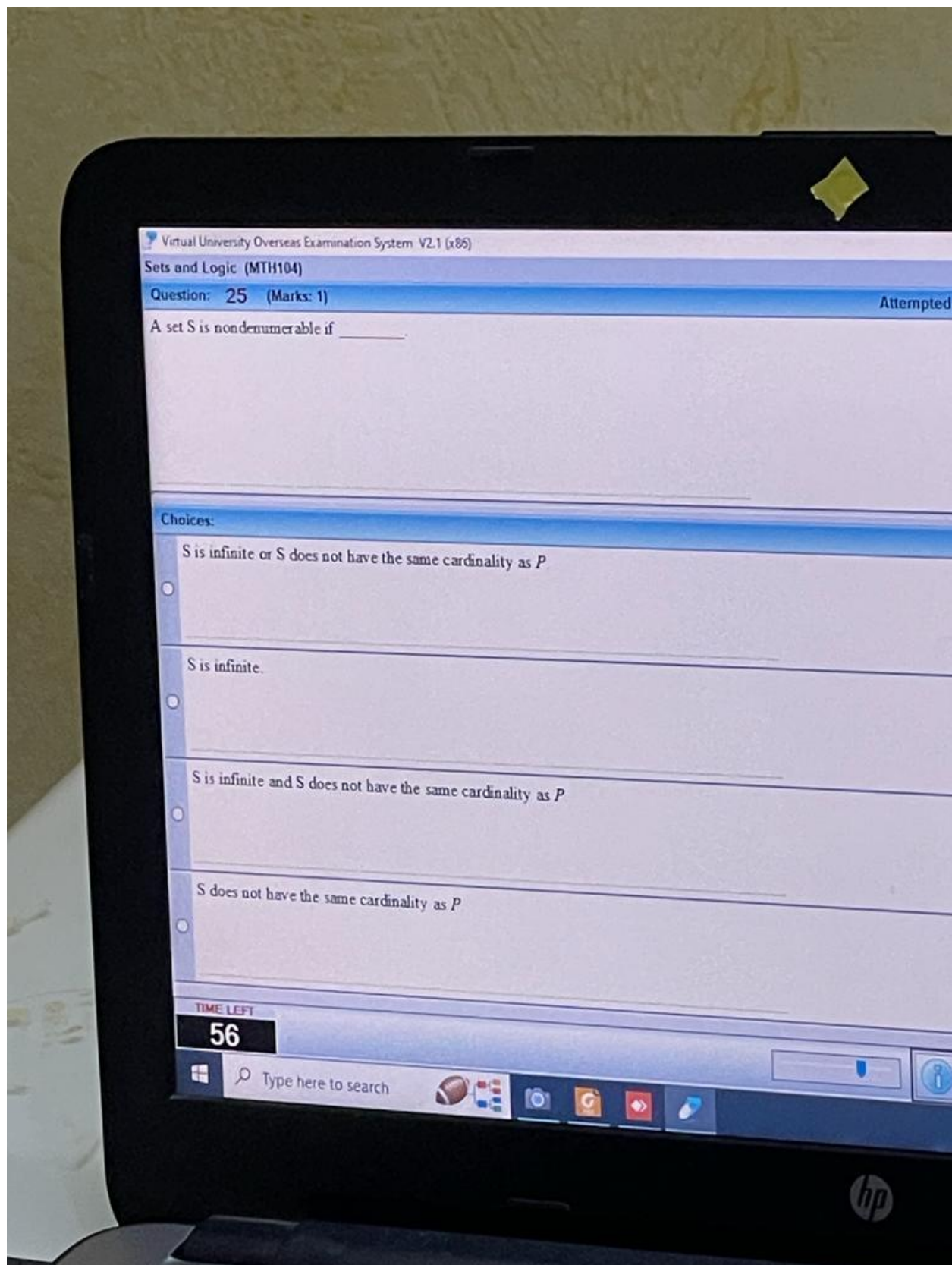
MTH104 MID TERM BY IQRA AZHAR



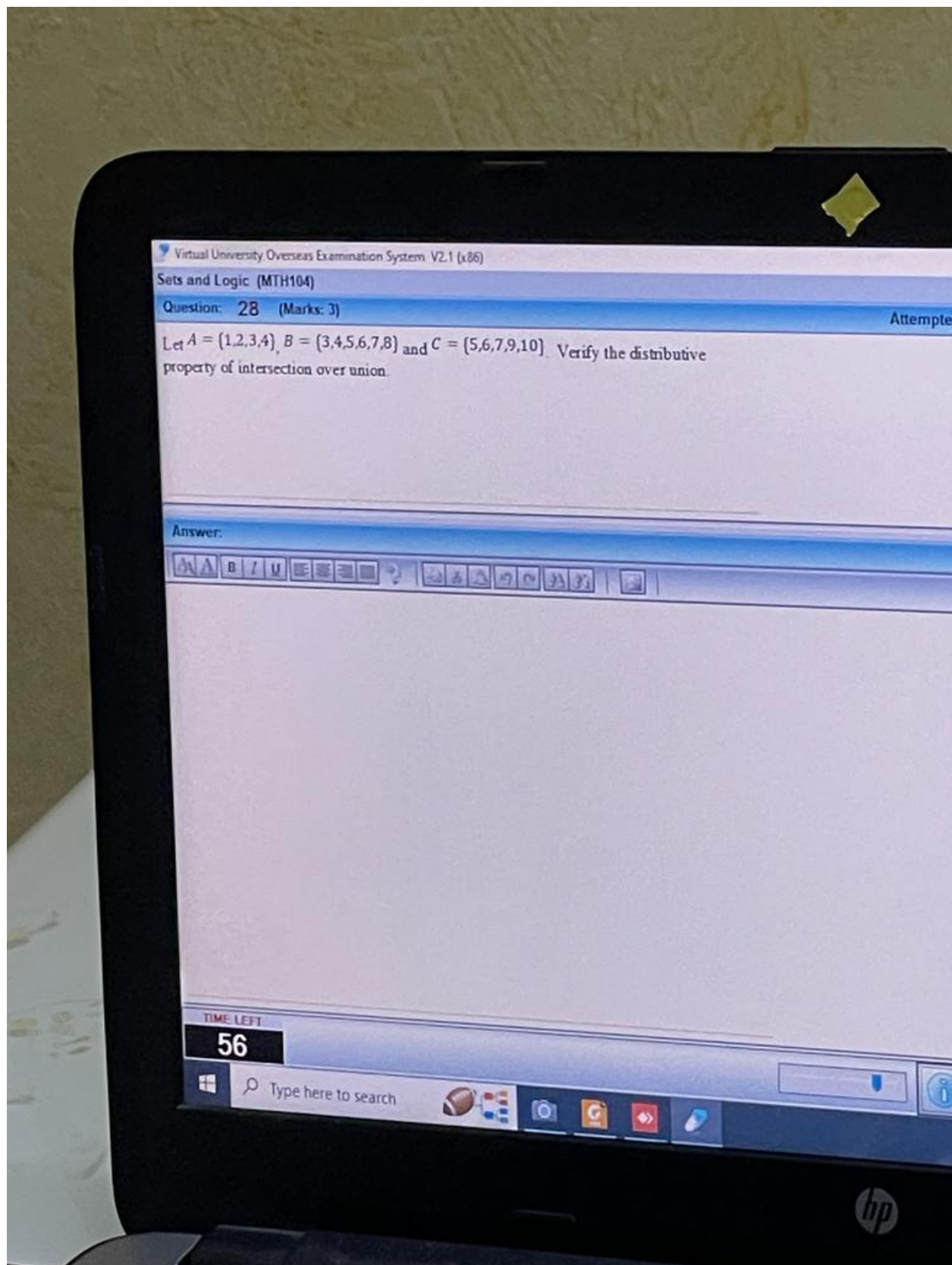
MTH104 MID TERM BY IQRA AZHAR



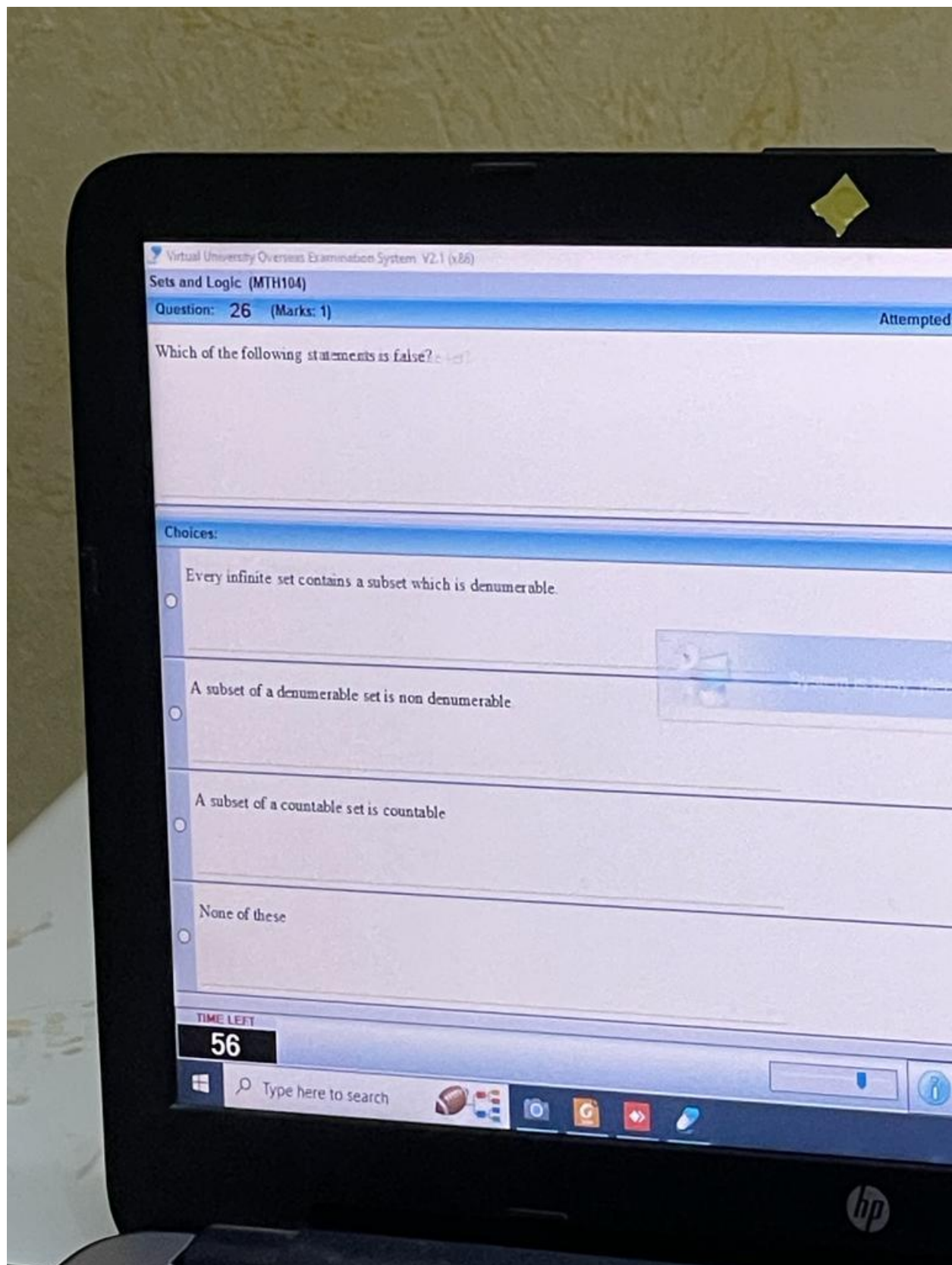
MTH104 MID TERM BY IQRA AZHAR



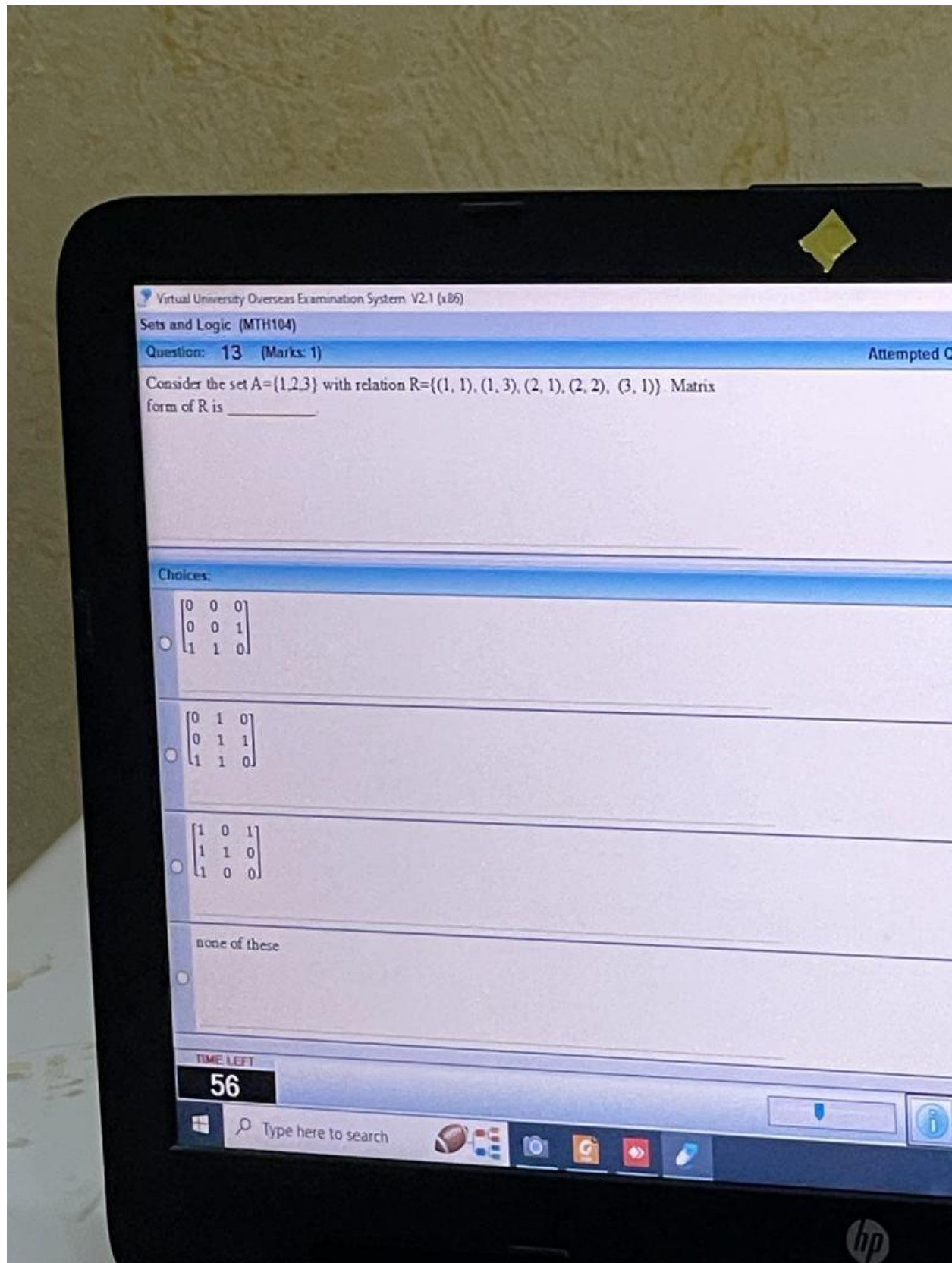
MTH104 MID TERM BY IQRA AZHAR



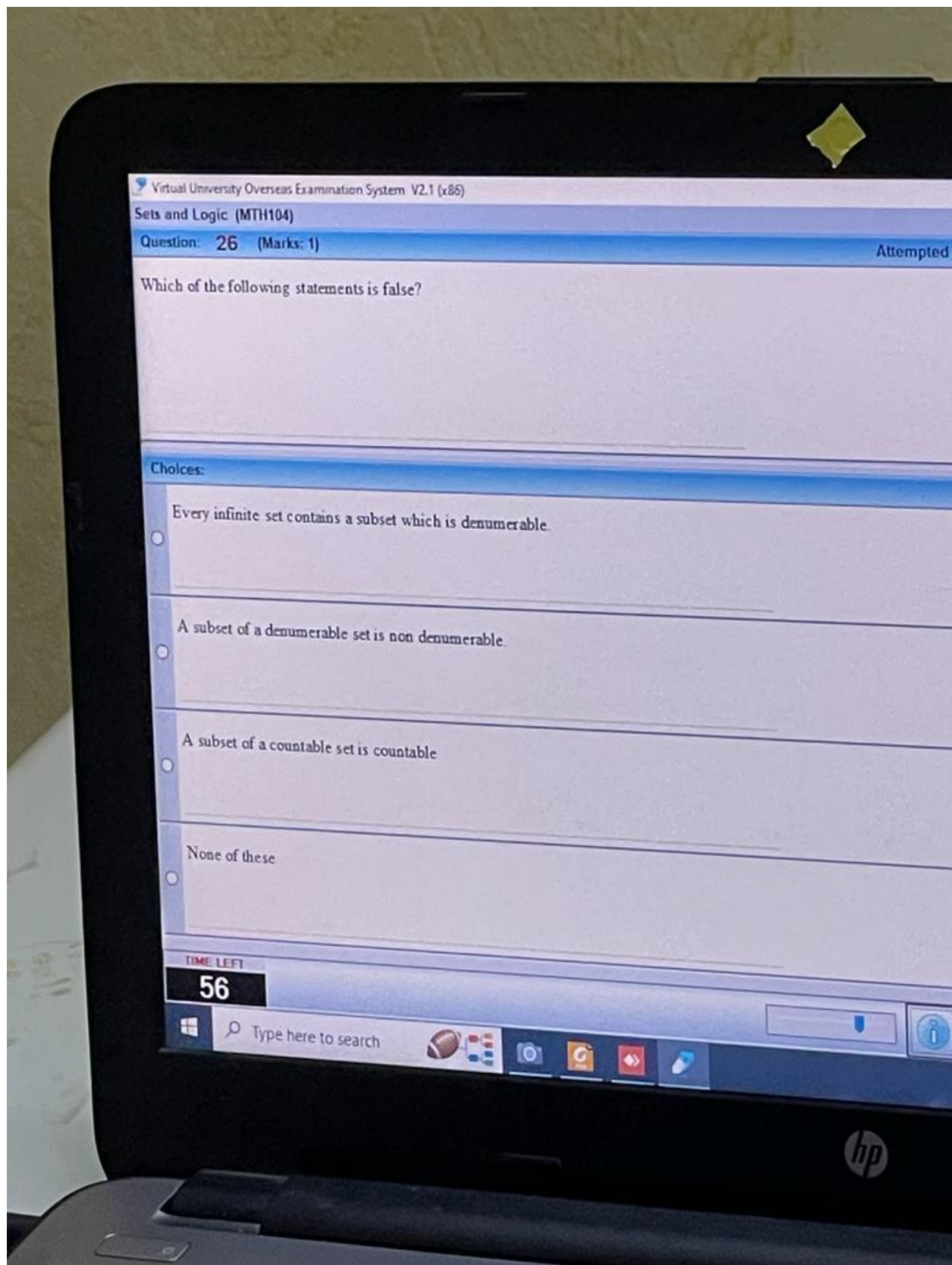
MTH104 MID TERM BY IQRA AZHAR



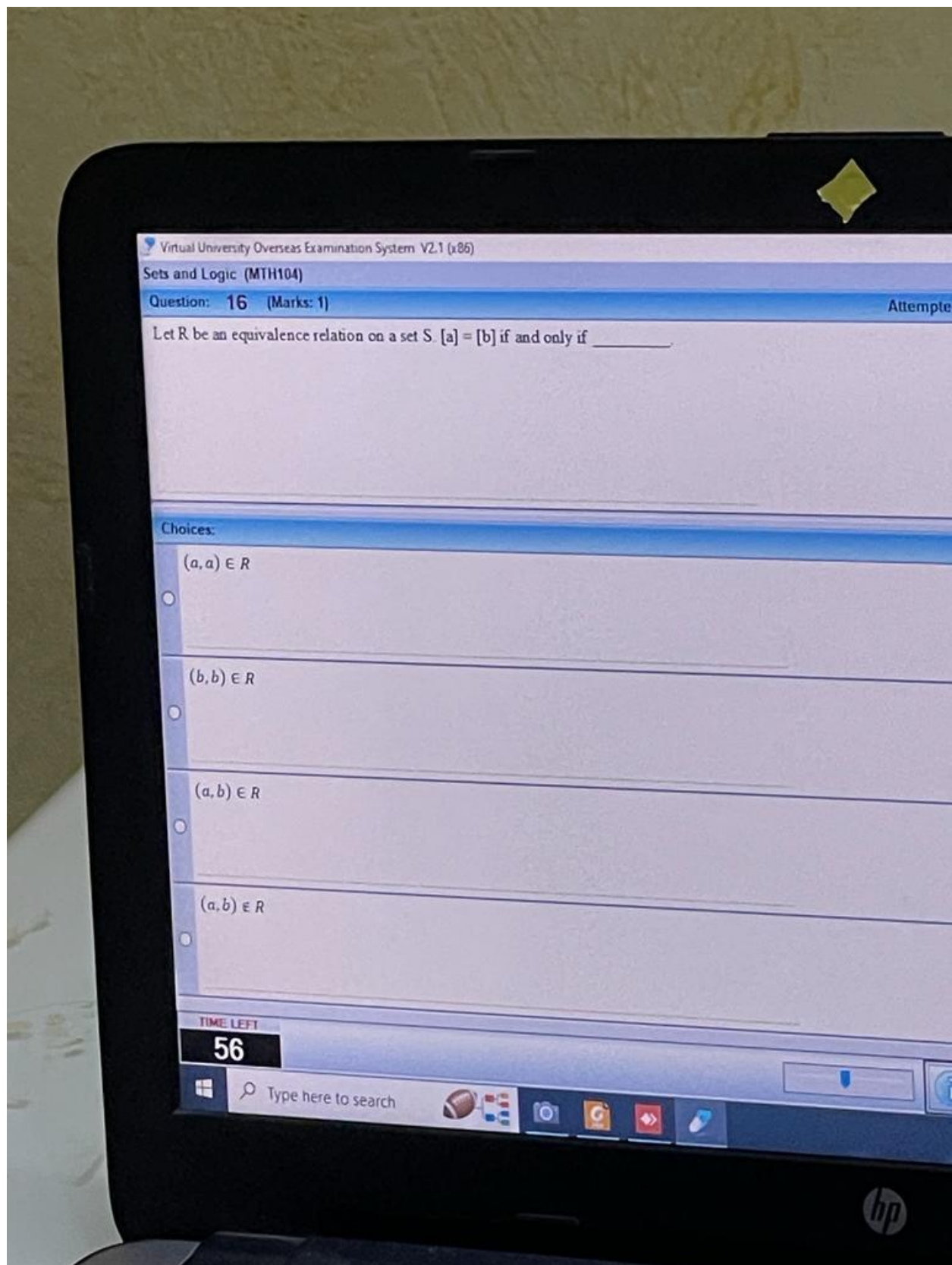
MTH104 MID TERM BY IQRA AZHAR



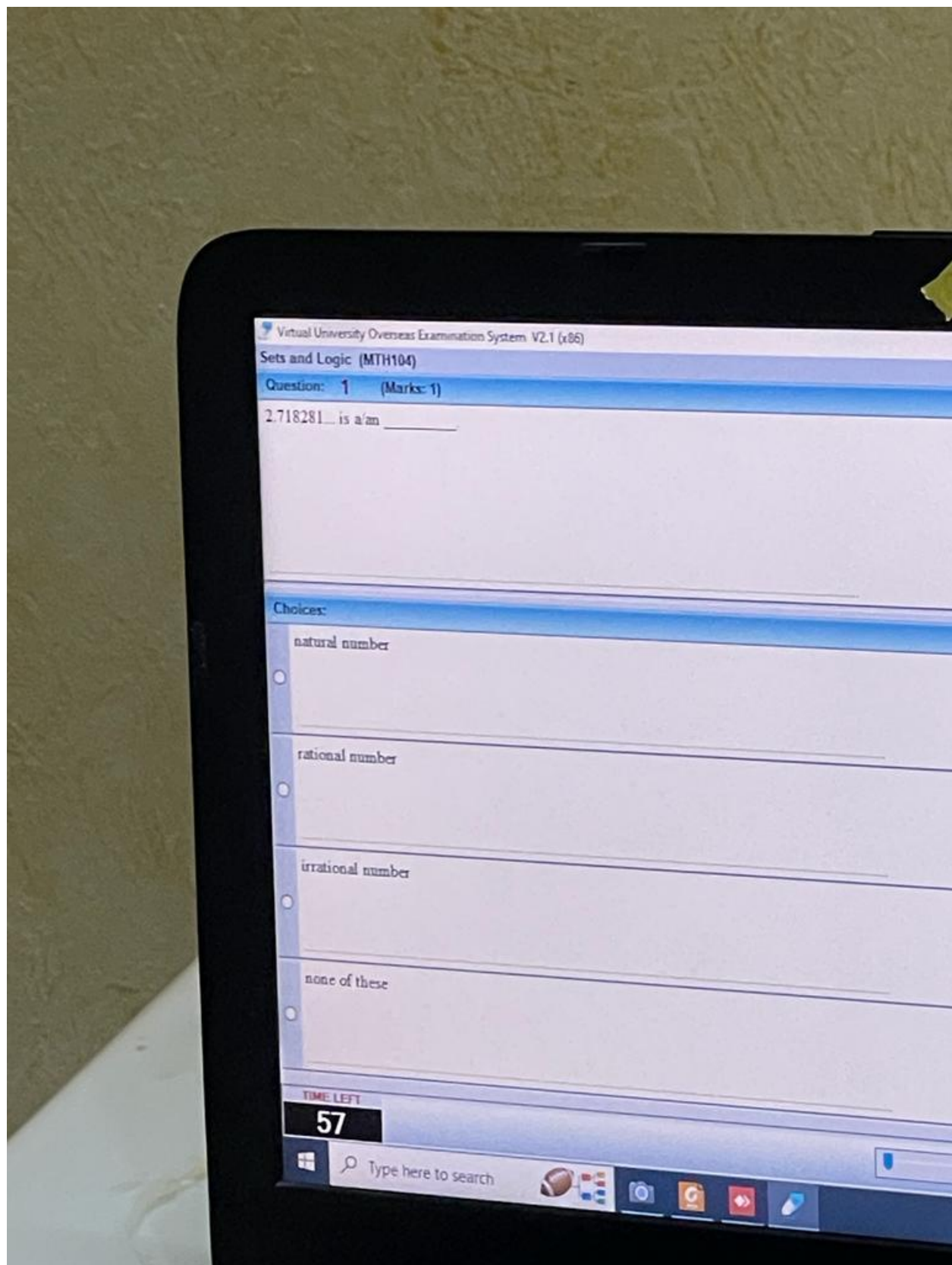
MTH104 MID TERM BY IQRA AZHAR



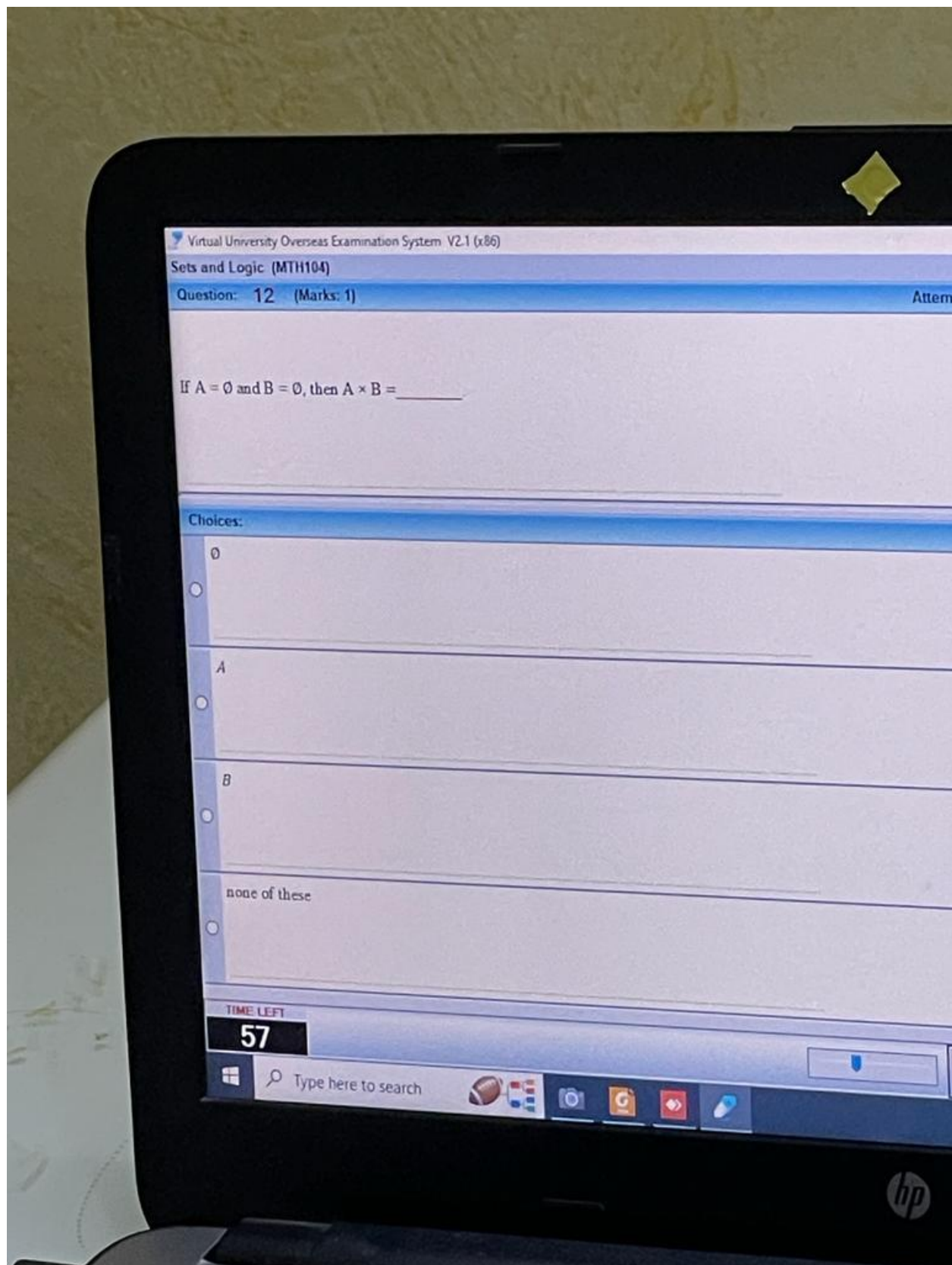
MTH104 MID TERM BY IQRA AZHAR



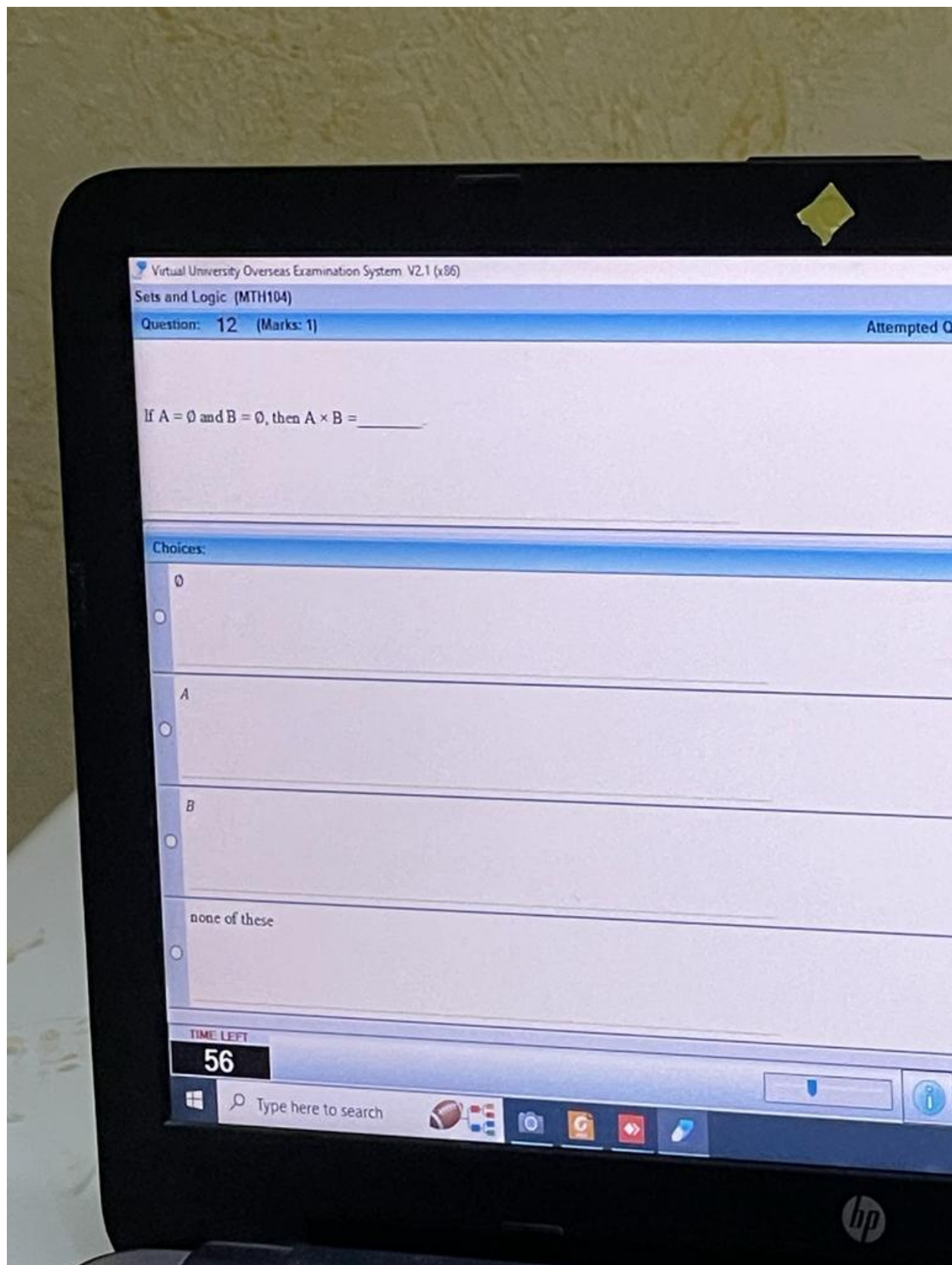
MTH104 MID TERM BY IQRA AZHAR



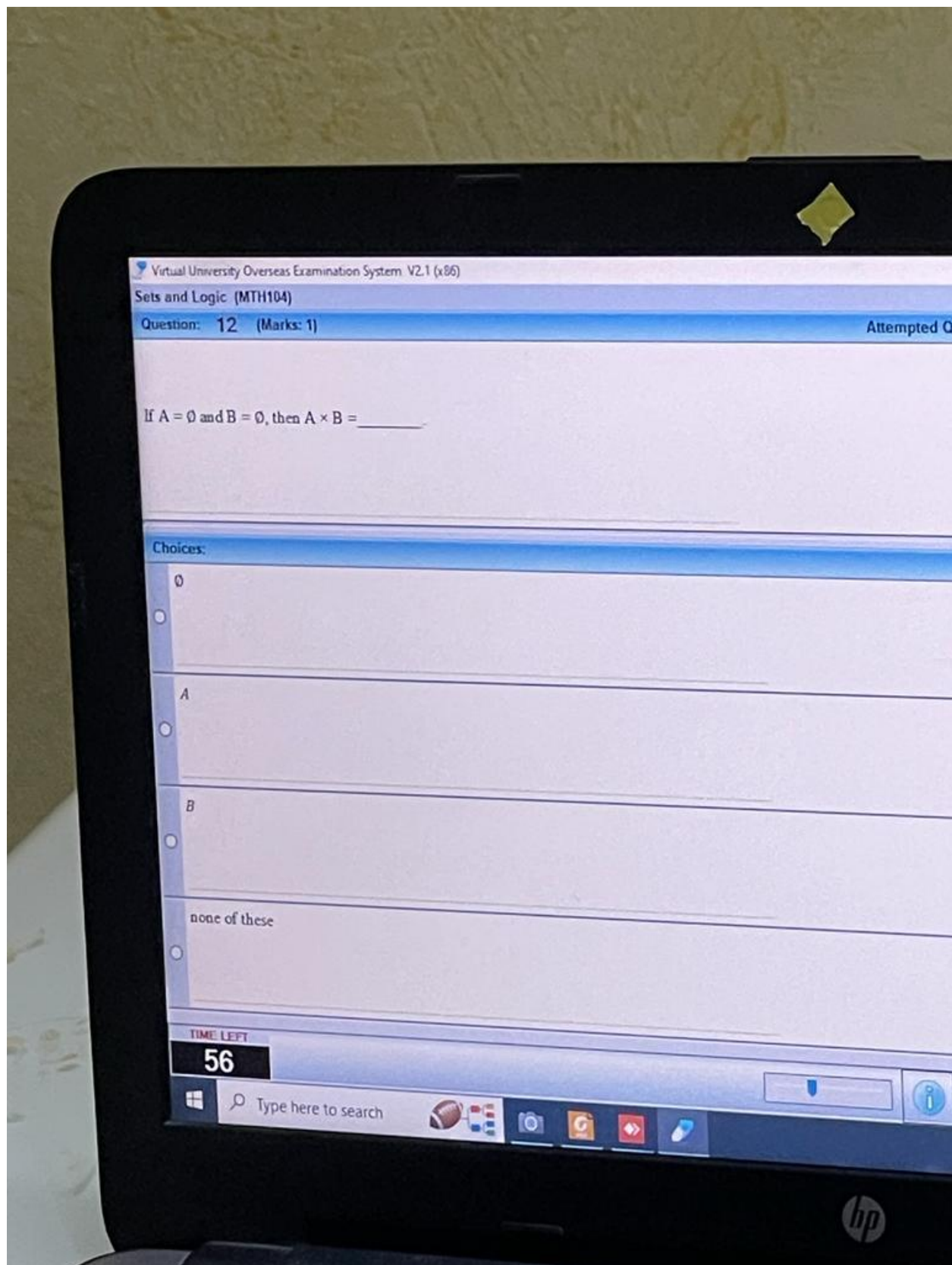
MTH104 MID TERM BY IQRA AZHAR



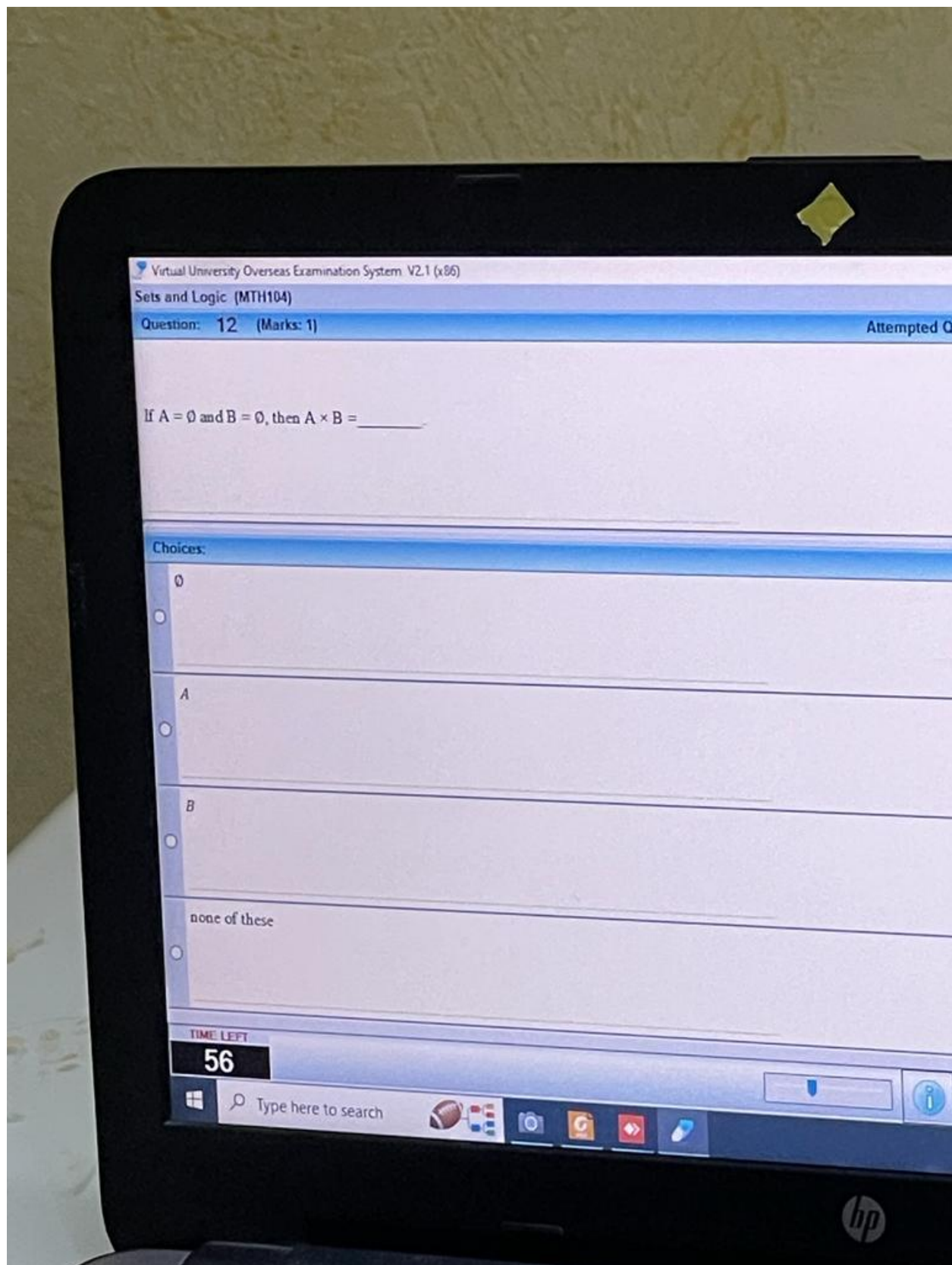
MTH104 MID TERM BY IQRA AZHAR



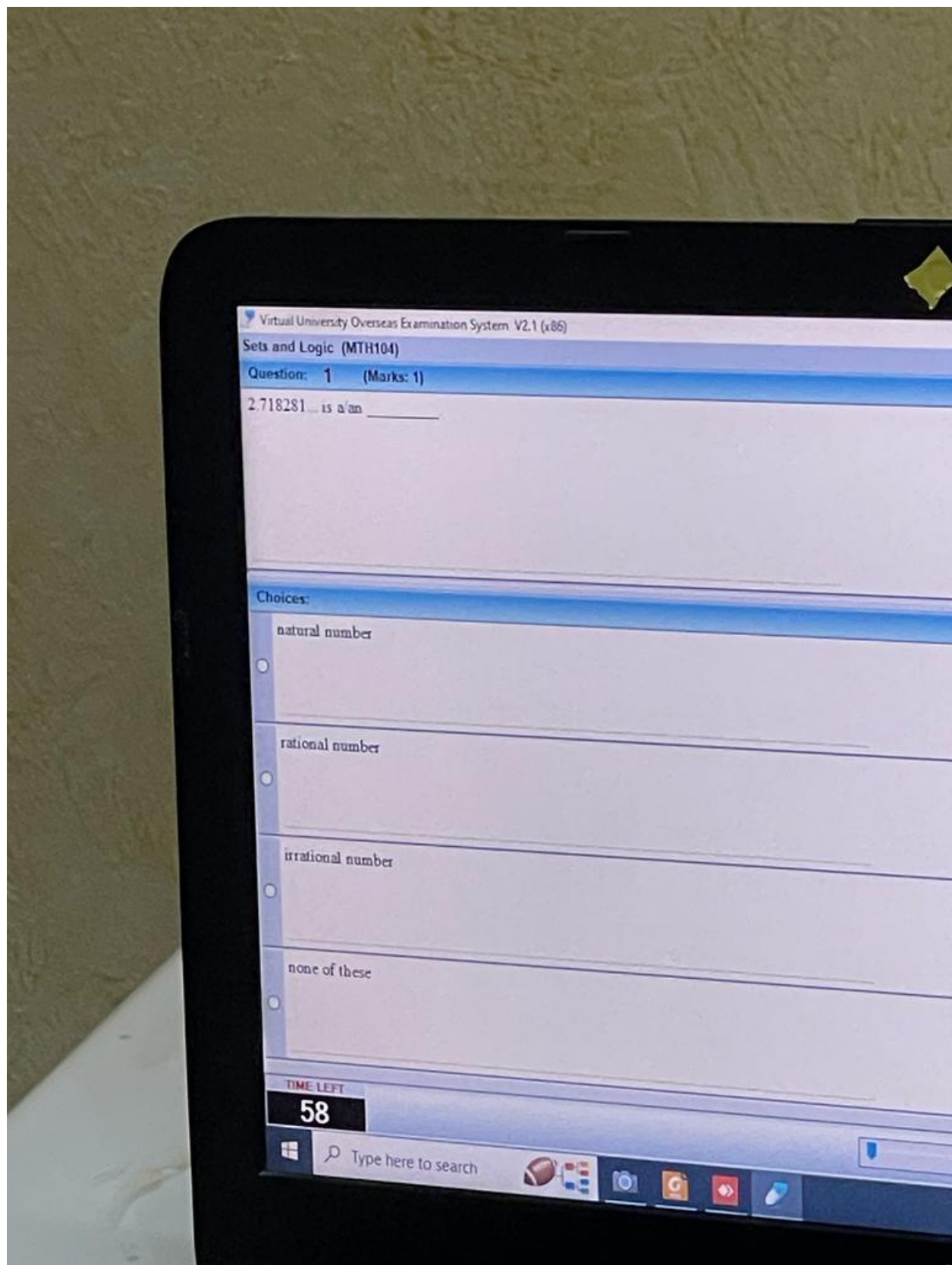
MTH104 MID TERM BY IQRA AZHAR



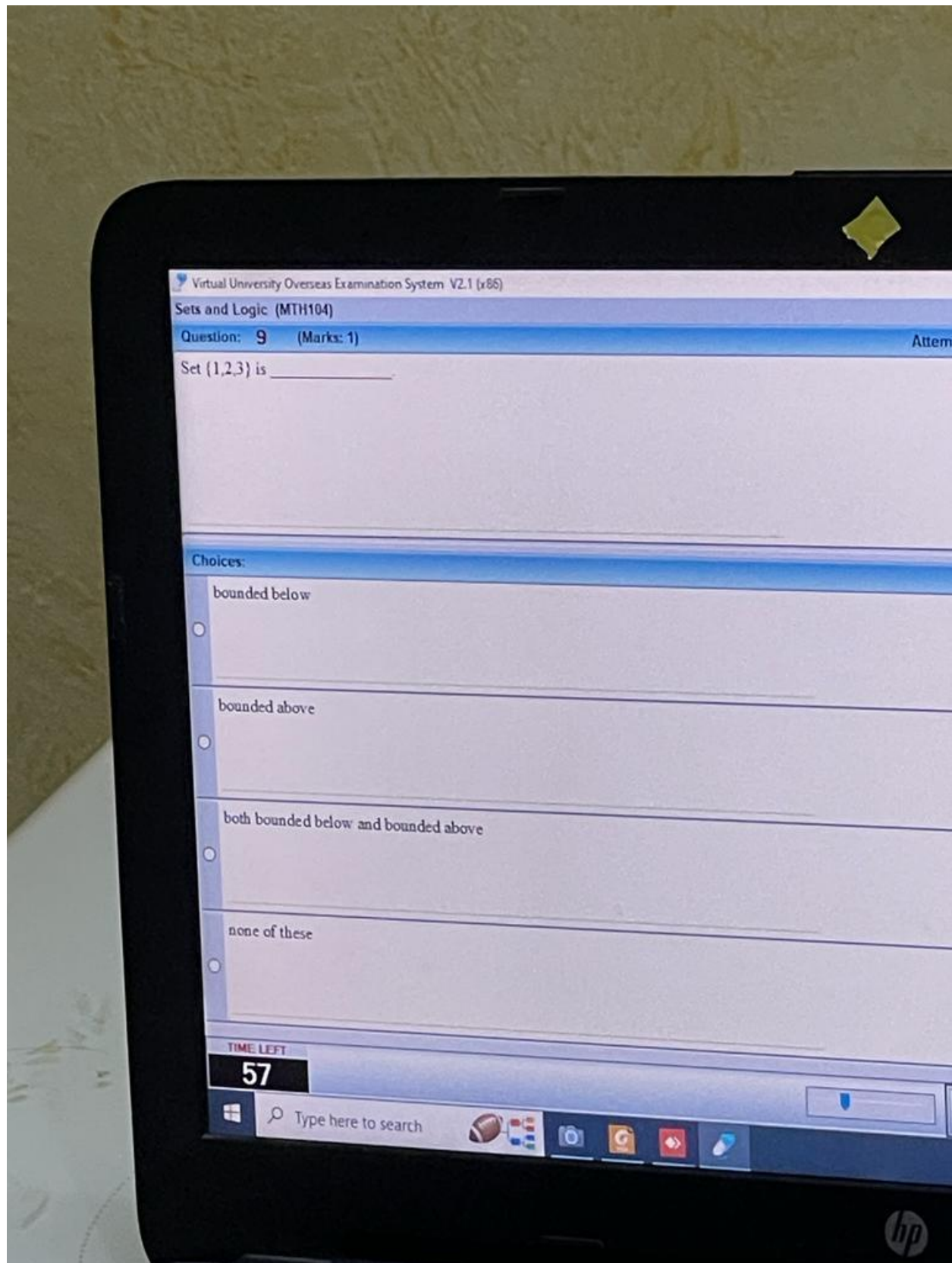
MTH104 MID TERM BY IQRA AZHAR



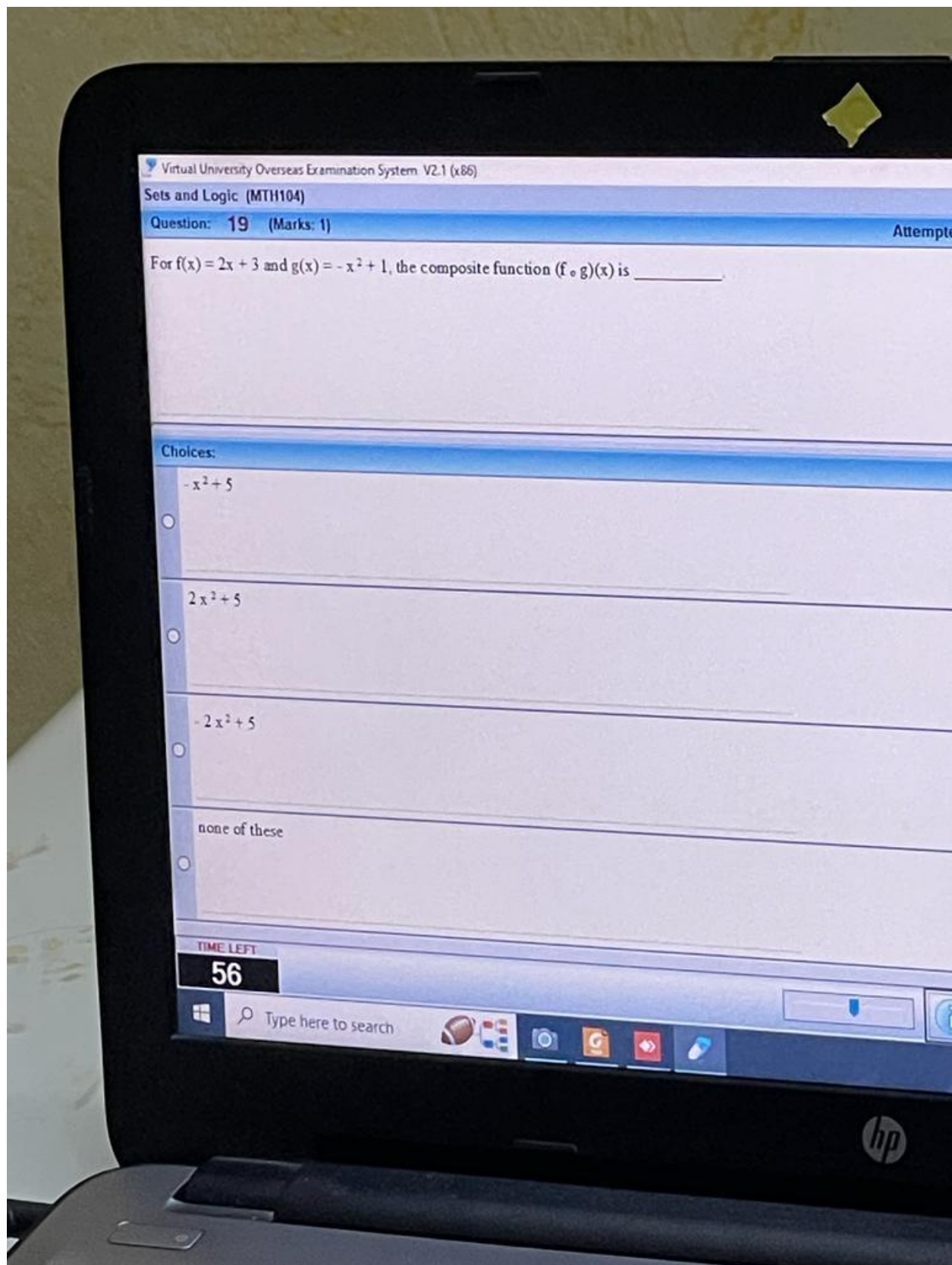
MTH104 MID TERM BY IQRA AZHAR



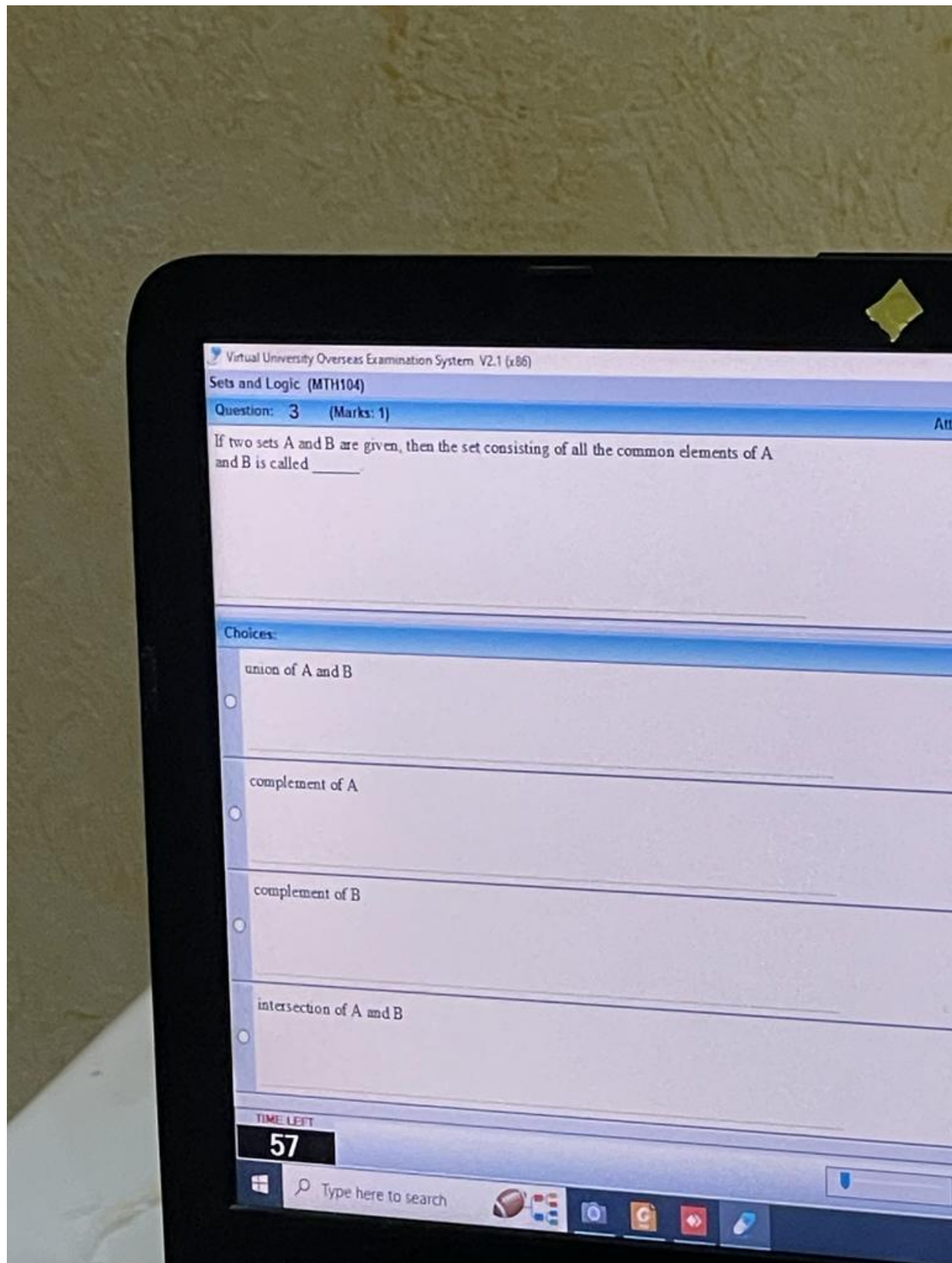
MTH104 MID TERM BY IQRA AZHAR



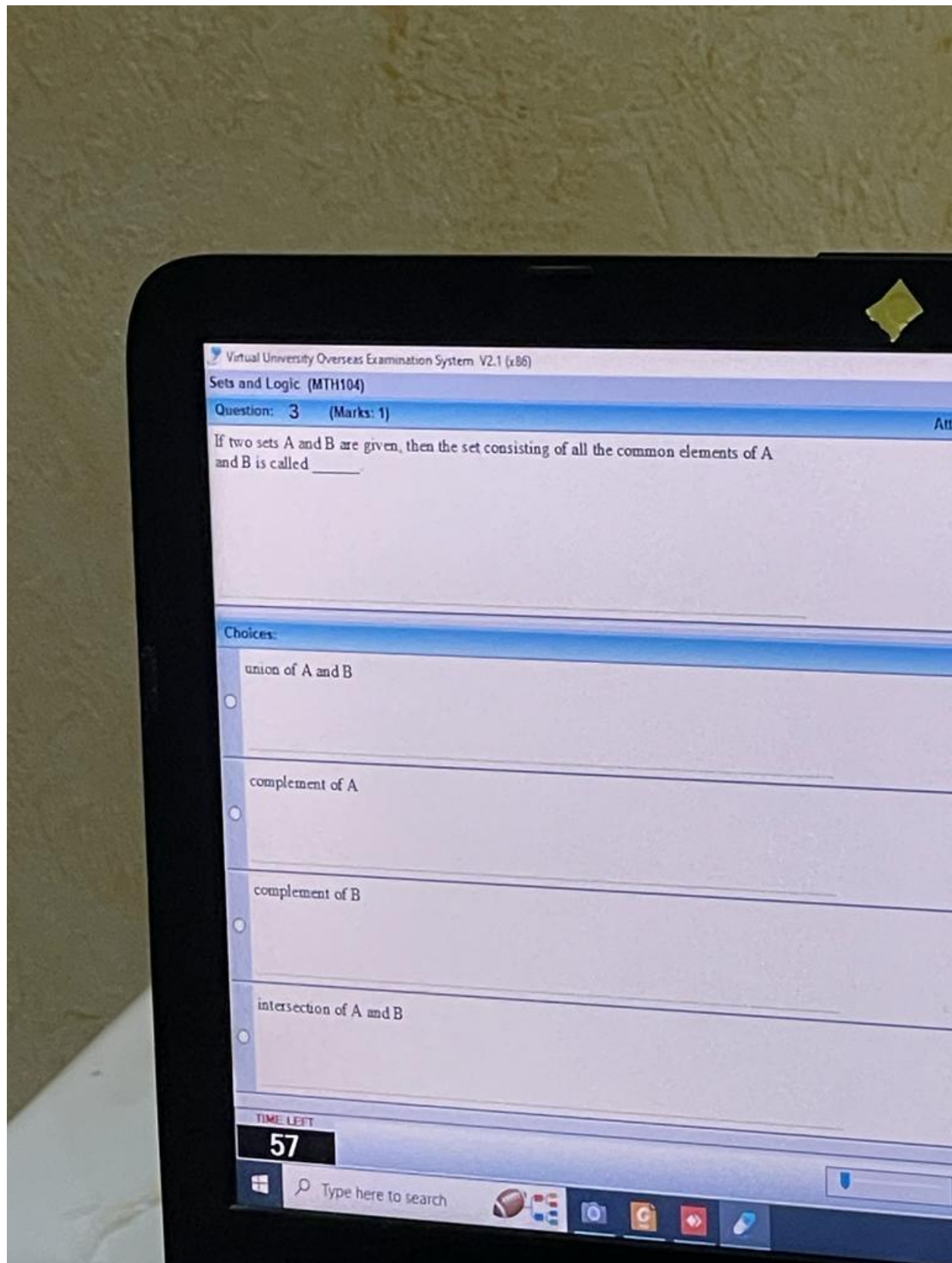
MTH104 MID TERM BY IQRA AZHAR



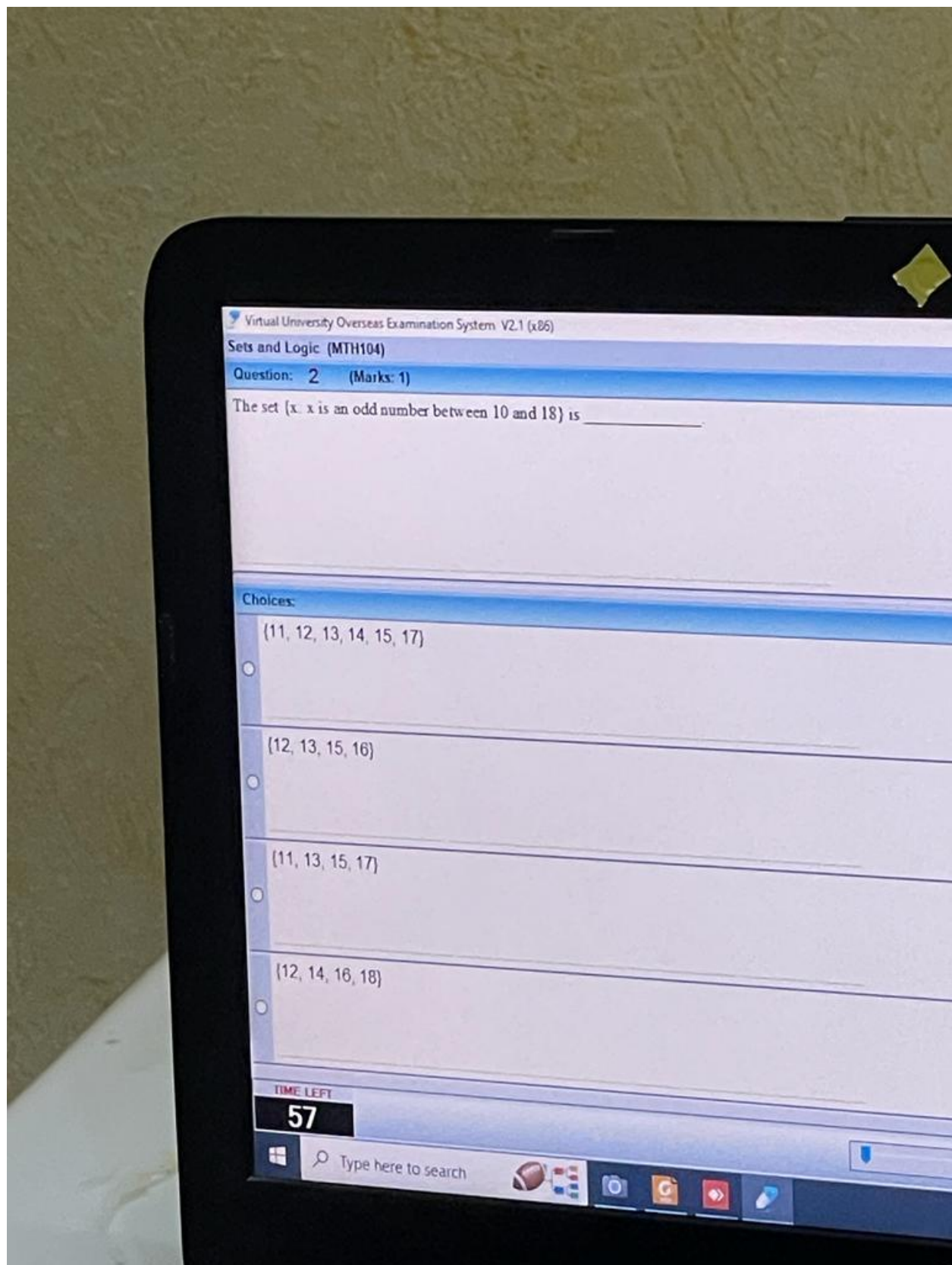
MTH104 MID TERM BY IQRA AZHAR



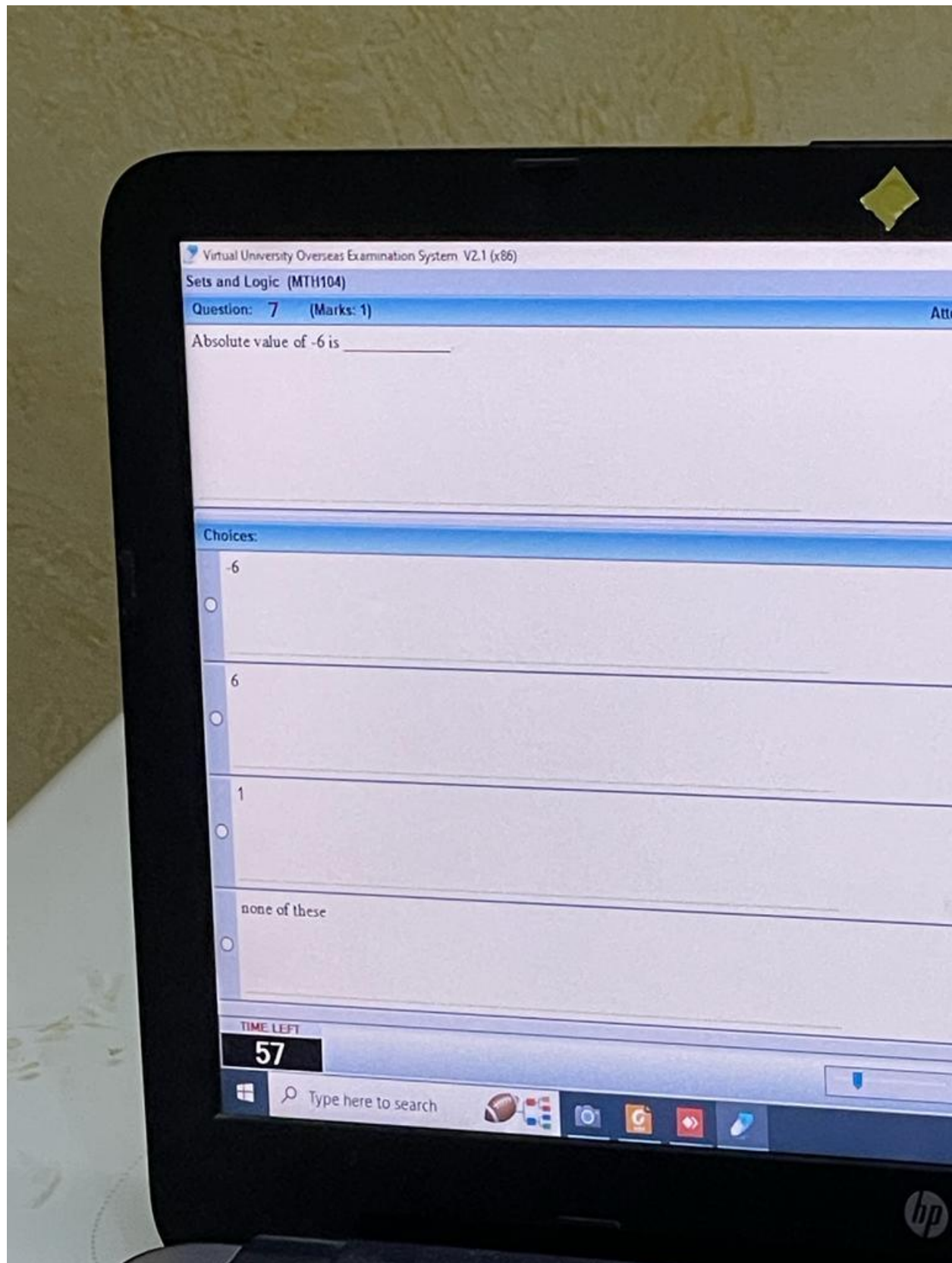
MTH104 MID TERM BY IQRA AZHAR



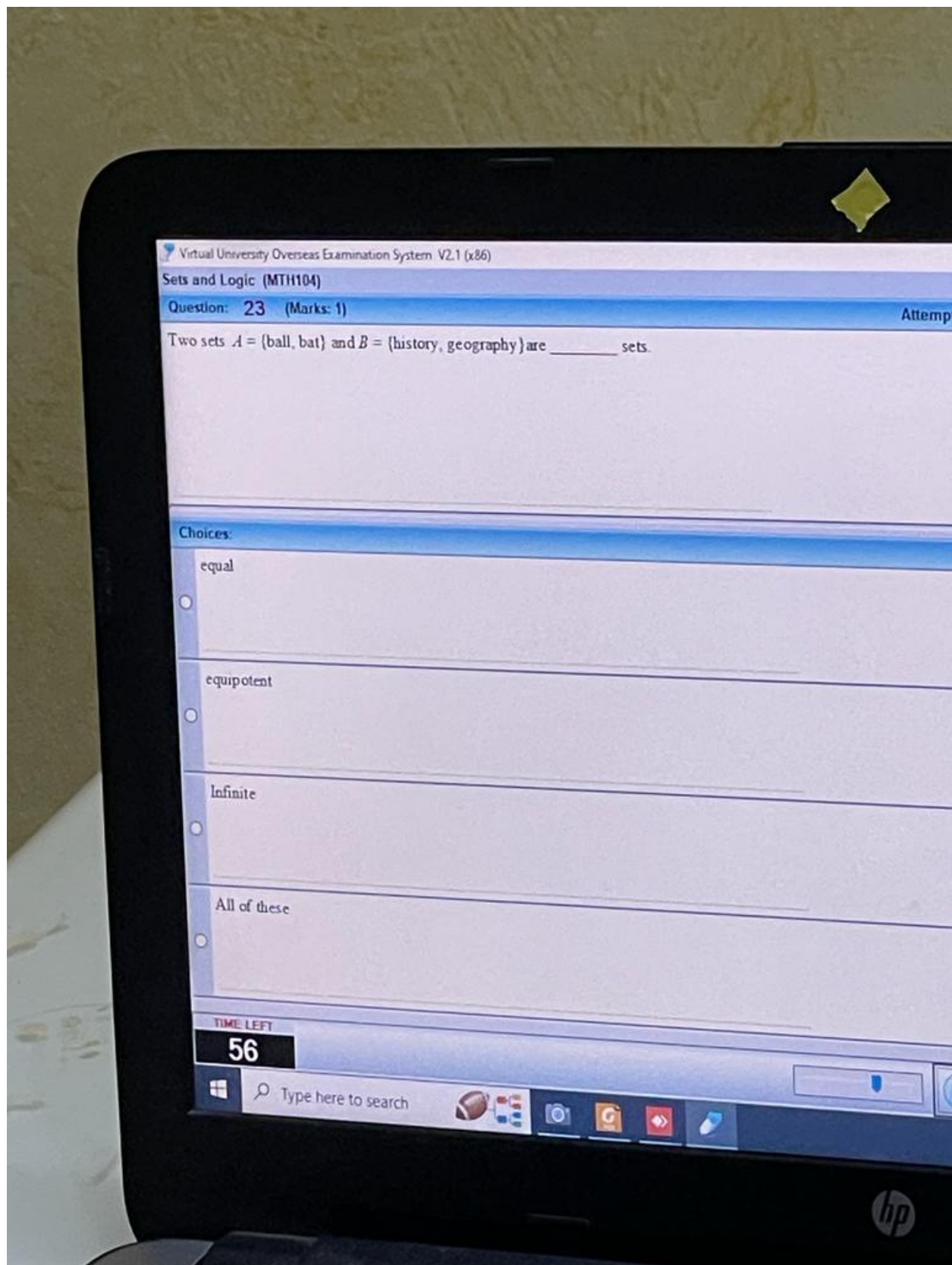
MTH104 MID TERM BY IQRA AZHAR



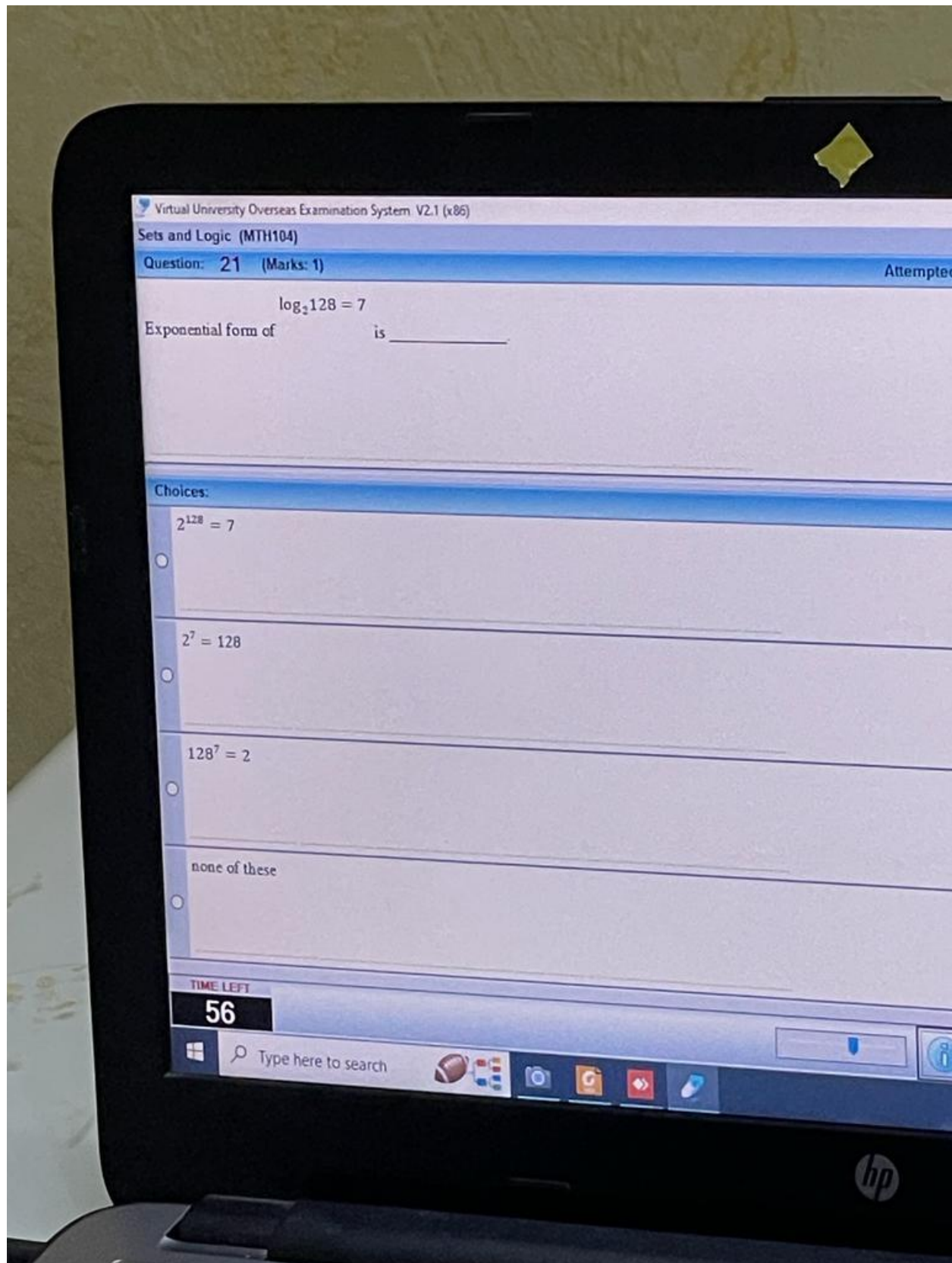
MTH104 MID TERM BY IQRA AZHAR



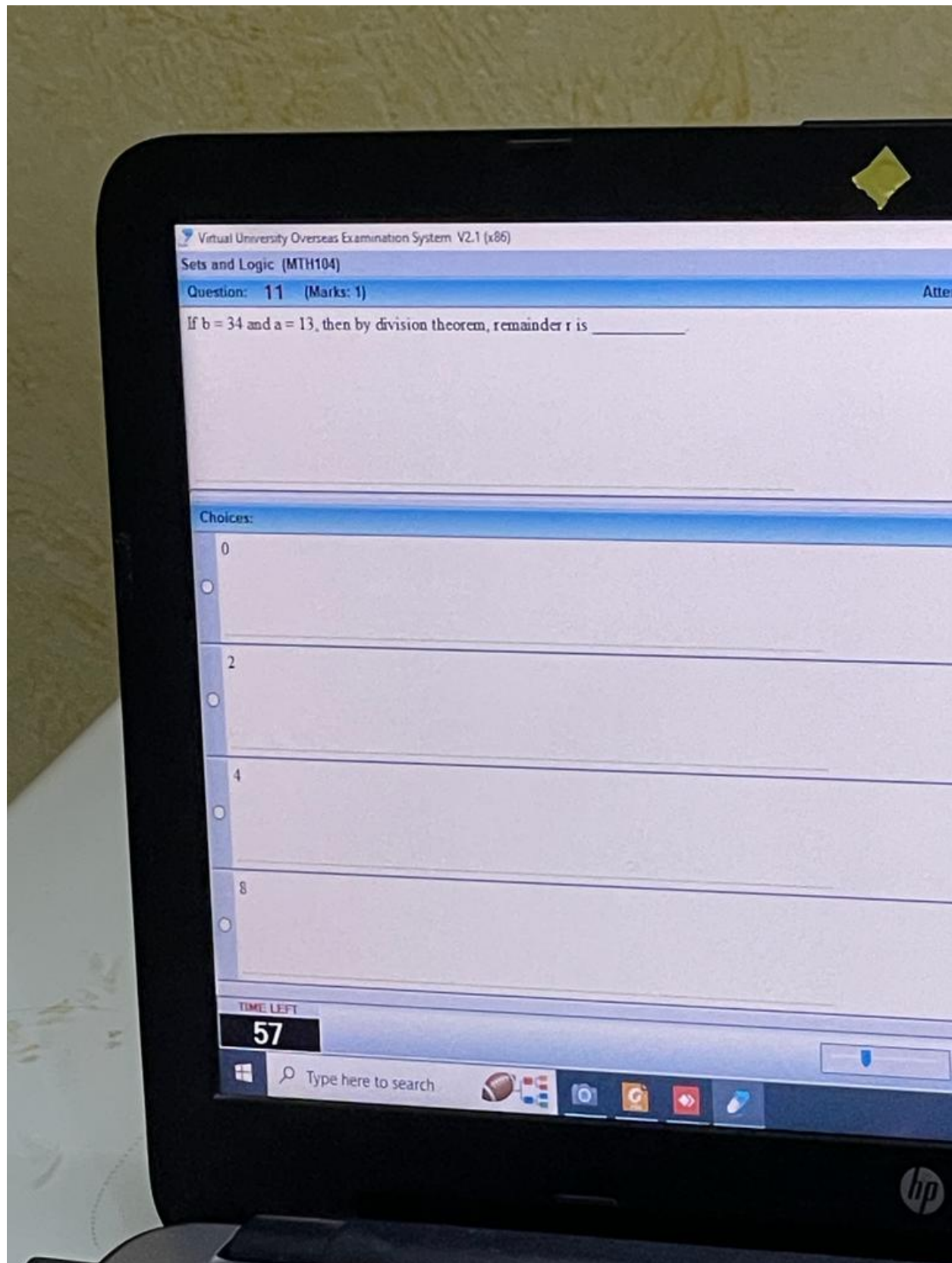
MTH104 MID TERM BY IQRA AZHAR



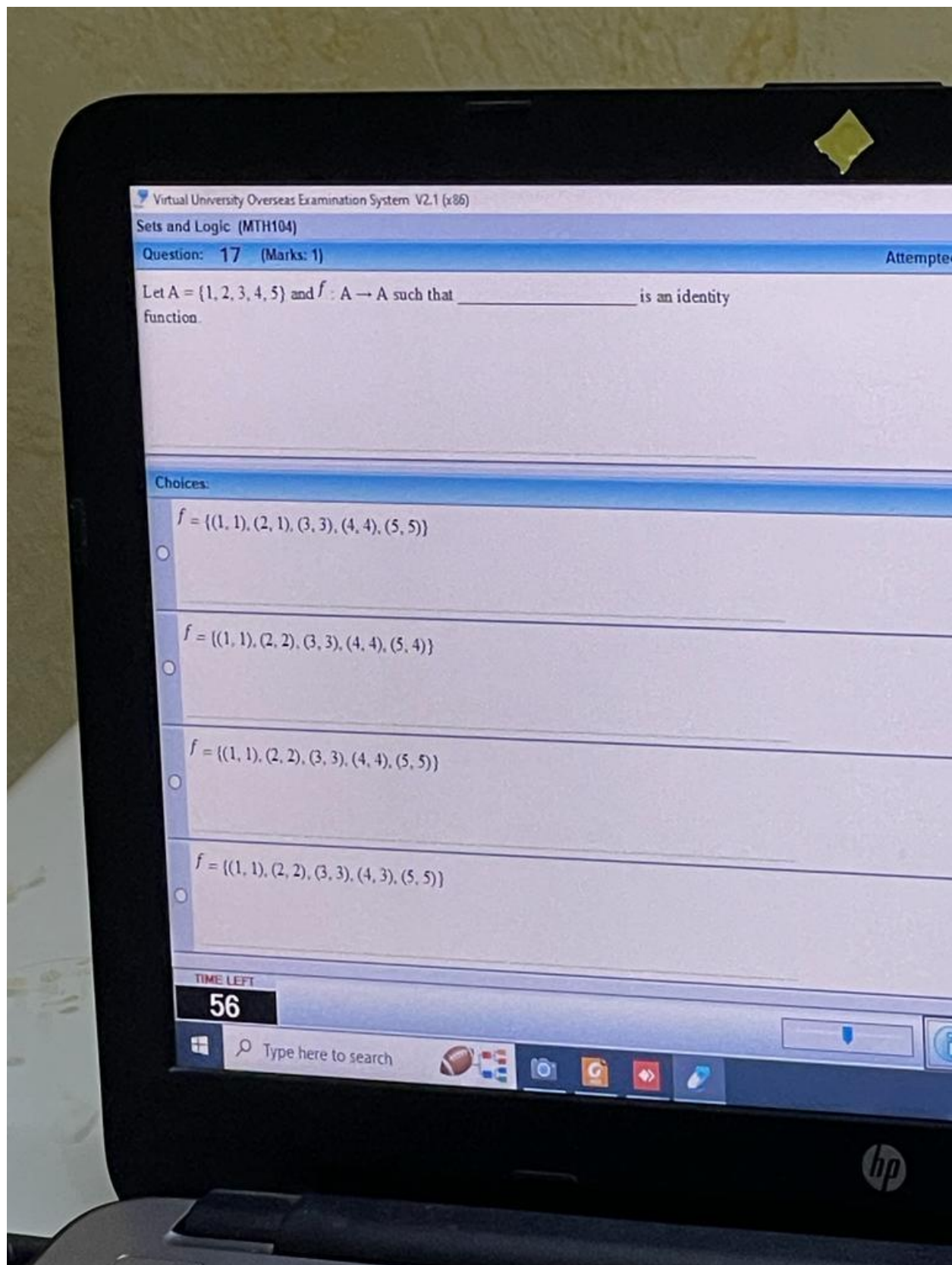
MTH104 MID TERM BY IQRA AZHAR



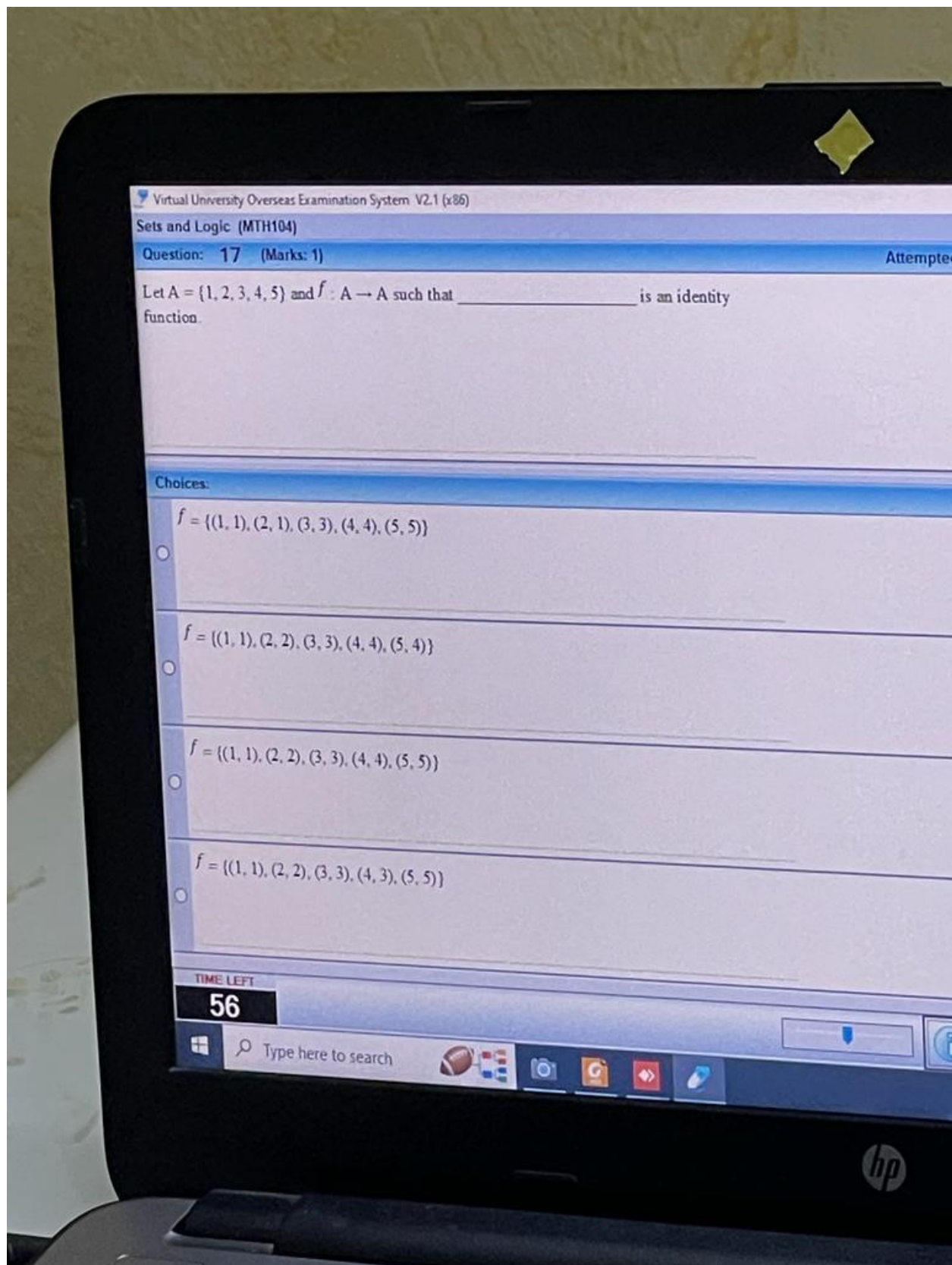
MTH104 MID TERM BY IQRA AZHAR



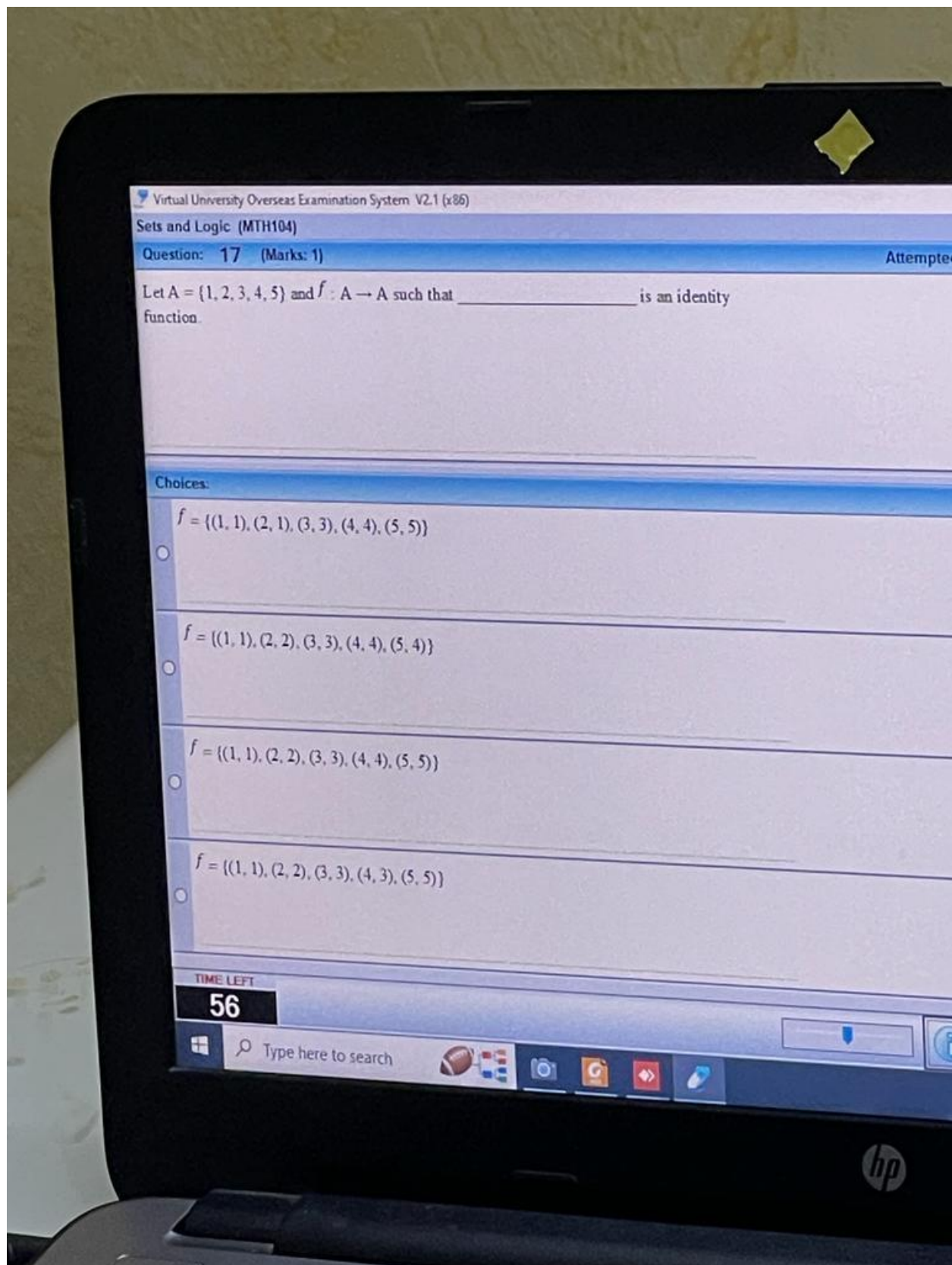
MTH104 MID TERM BY IQRA AZHAR



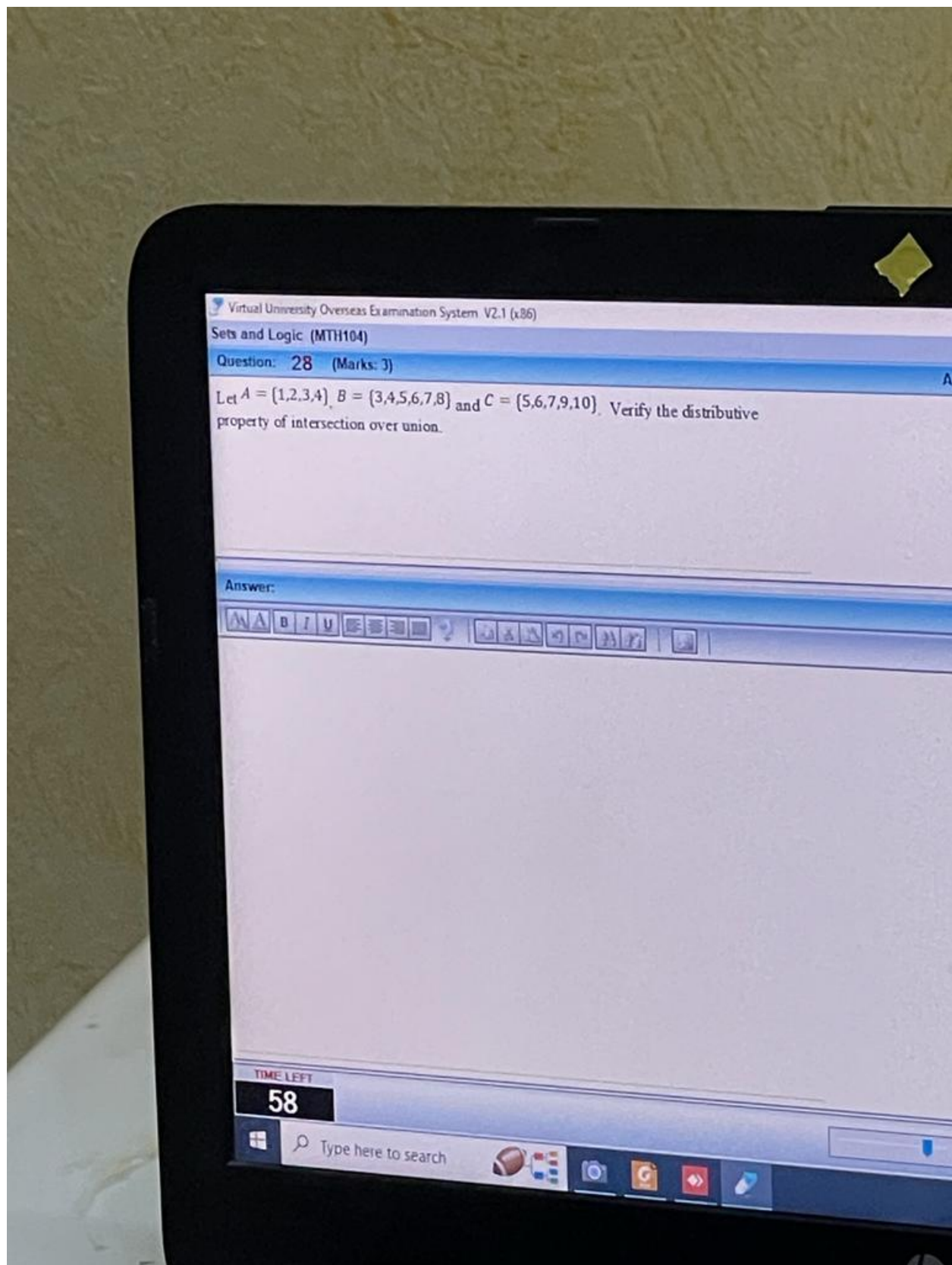
MTH104 MID TERM BY IQRA AZHAR



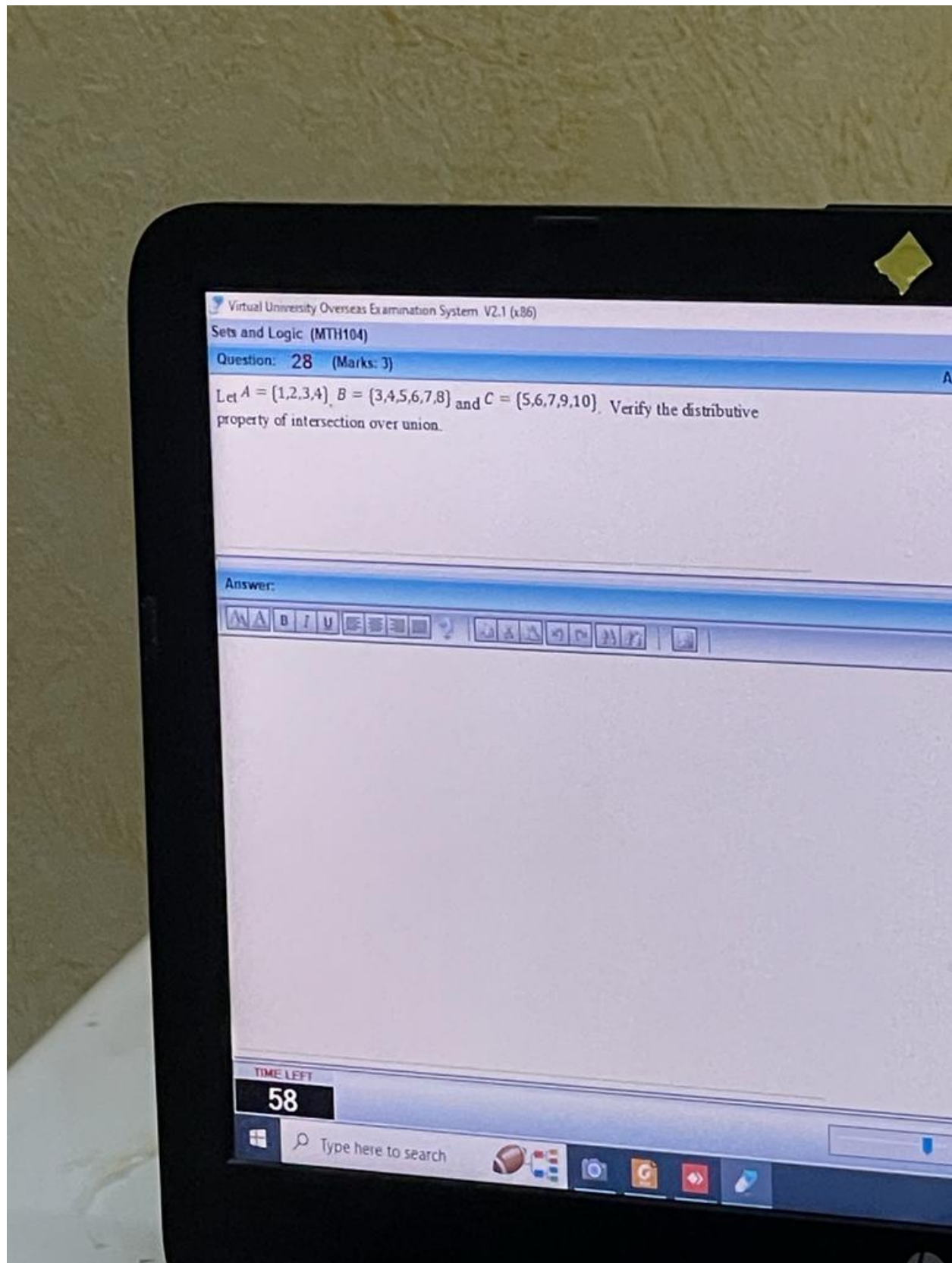
MTH104 MID TERM BY IQRA AZHAR



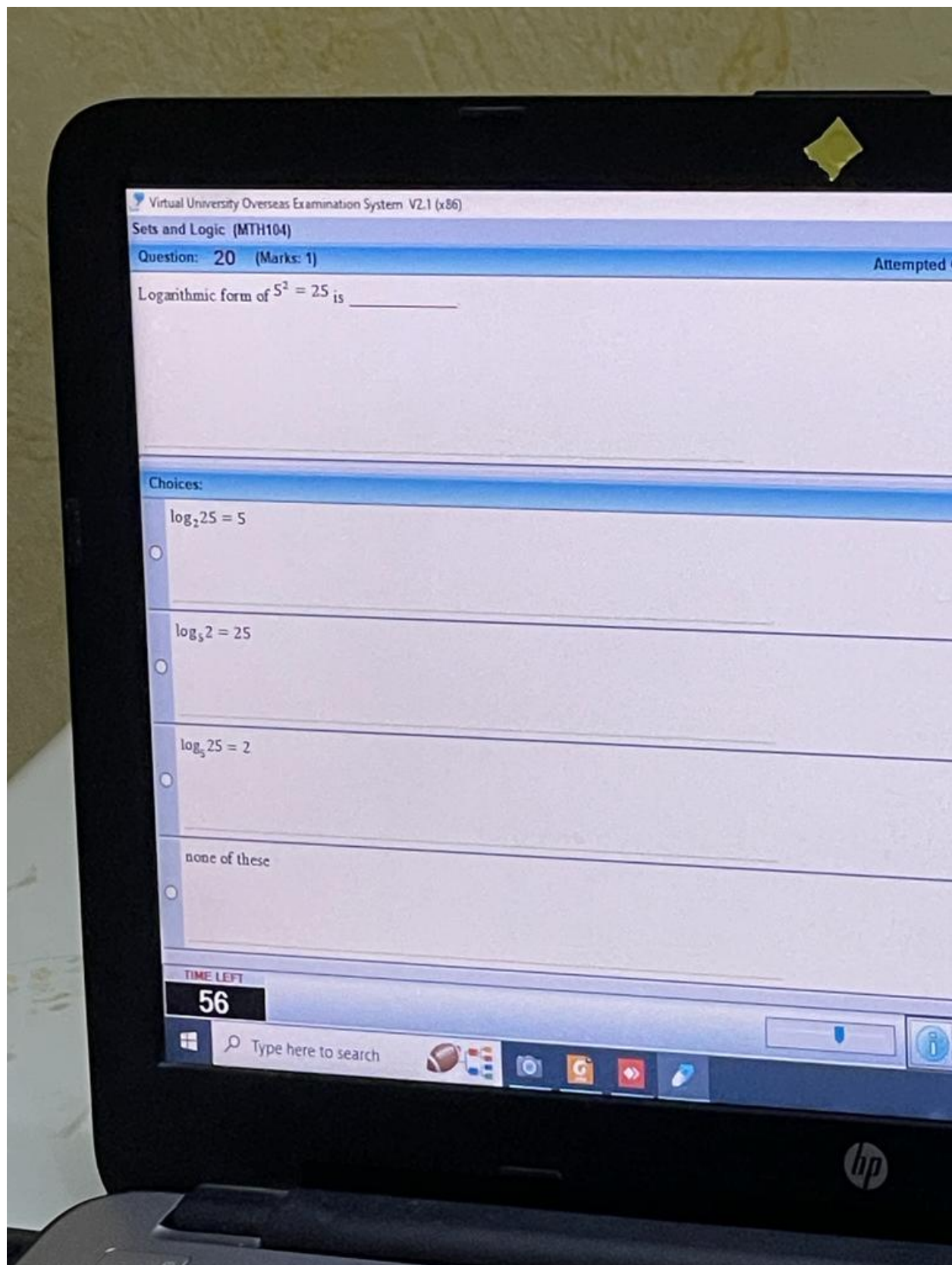
MTH104 MID TERM BY IQRA AZHAR



MTH104 MID TERM BY IQRA AZHAR

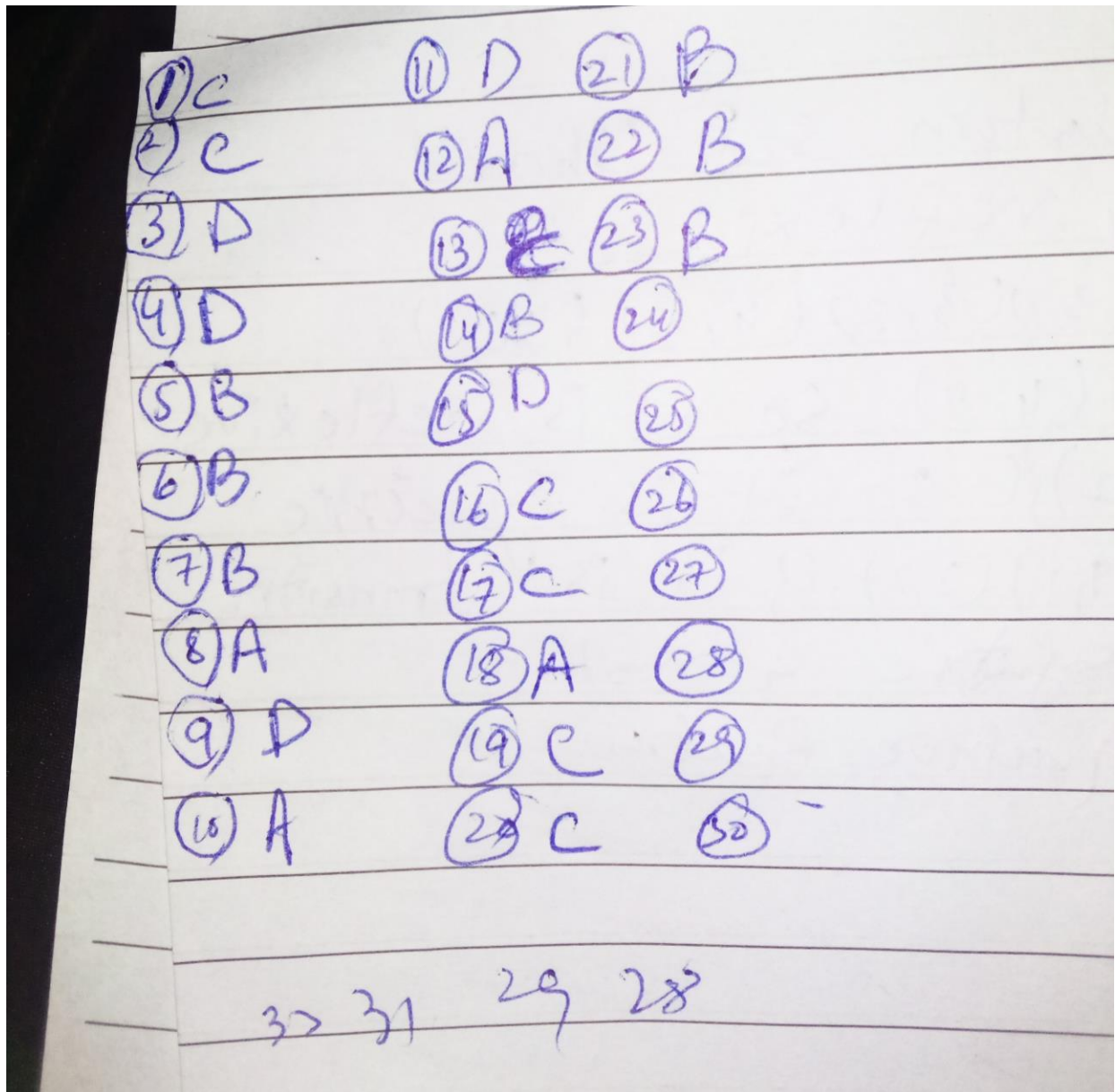


MTH104 MID TERM BY IQRA AZHAR



MTH104 MID TERM BY IQRA AZHAR

Answer



Q28

MTH104 MID TERM BY IQRA AZHAR

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Let's verify this property using the given sets:

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6, 7, 8\}$$

$$C = \{5, 6, 7, 9, 10\}$$

1. **Left-hand side (LHS):** $A \cap (B \cup C)$

$$A \cap (B \cup C) = \{1, 2, 3, 4\} \cap \{3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A \cap (B \cup C) = \{3, 4\}$$

2. **Right-hand side (RHS):** $(A \cap B) \cup (A \cap C)$

$$(A \cap B) \cup (A \cap C) = (\{1, 2, 3, 4\} \cap \{3, 4, 5, 6, 7, 8\}) \cup (\{1, 2, 3, 4\} \cap \{5, 6, 7, 9, 10\})$$

$$(A \cap B) \cup (A \cap C) = (\{3, 4\}) \cup (\{3, 4\})$$

$$(A \cap B) \cup (A \cap C) = \{3, 4\}$$

As we can see, the left-hand side (LHS) is equal to the right-hand side (RHS). Therefore, the distributive property of intersection over union is verified for the given sets A, B, and C.

MTH104 MID TERM BY IQRA AZHAR

Let's use the Euclidean algorithm to find the GCD of 124 and 348:

Step 1: Divide 348 by 124.

$$348 = 2 \times 124 + 100$$

Step 2: Replace 348 with 124 and 124 with the remainder (100).

$$124 = 1 \times 100 + 24$$

Step 3: Replace 124 with 100 and 100 with the remainder (24).

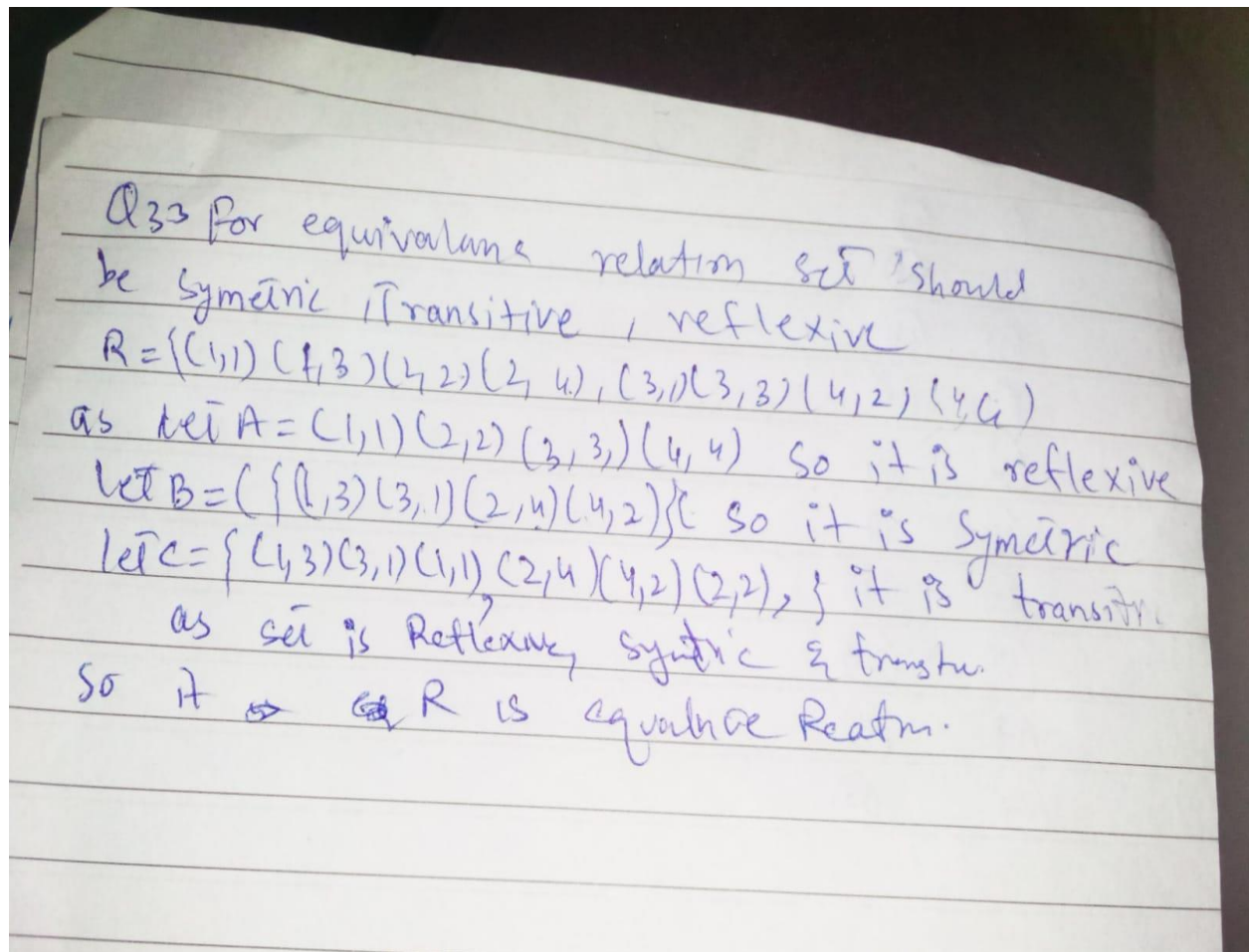
$$100 = 4 \times 24 + 4$$

Step 4: Replace 100 with 24 and 24 with the remainder (4).

$$24 = 6 \times 4 + 0$$

The remainder is now zero, so the last non-zero remainder (4) is the GCD of 124 and 348.

MTH104 MID TERM BY IQRA AZHAR



Q31

Q29